

ESAT PRACTICE TEST SERIES

MATHS 2 + CHEMISTRY

LEVEL 2 • HARDER THAN ESAT

Maths 2 + Chemistry

Mathematics 1 + Mathematics 2 + Chemistry

Challenge material with denser interpretation, quantitative reasoning and longer solution chains.

MODULE

Mathematics 1

MODULE

Mathematics 2

MODULE

Chemistry

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions • three verified module keys

Maths 2 + Chemistry · Level 2 Harder than ESAT

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Mathematics 2 + Chemistry. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 5 · preserved Mathematics 1 module

Q1 C	Q2 B	Q3 D	Q4 A	Q5 E
Q6 C	Q7 B	Q8 D	Q9 A	Q10 E
Q11 C	Q12 B	Q13 D	Q14 A	Q15 E
Q16 B	Q17 C	Q18 D	Q19 A	Q20 E
Q21 C	Q22 B	Q23 D	Q24 A	Q25 E
Q26 C	Q27 B			

Mathematics 2

Engineering Mock Test 5 · preserved Mathematics 2 module

Q1 D	Q2 B	Q3 E	Q4 C	Q5 B
Q6 C	Q7 B	Q8 B	Q9 C	Q10 B
Q11 B	Q12 B	Q13 C	Q14 B	Q15 B
Q16 C	Q17 C	Q18 D	Q19 D	Q20 C
Q21 C	Q22 B	Q23 C	Q24 C	Q25 B
Q26 E	Q27 D			

Chemistry

Preserved Chemistry Harder than ESAT source set

Q1 A	Q2 D	Q3 F	Q4 H	Q5 C
Q6 G	Q7 B	Q8 G	Q9 E	Q10 F
Q11 C	Q12 B	Q13 E	Q14 A	Q15 C
Q16 D	Q17 C	Q18 B	Q19 E	Q20 B
Q21 A	Q22 D	Q23 F	Q24 D	Q25 F
Q26 E	Q27 A			

Mathematics 1

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 5 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Laws of indices and exact simplification

Answer C**QUESTION 1**

Simplify fully

$$\frac{(50x^{3/2}y^{-1})^2(2x^{-1}y^{5/2})^3}{(20x^{1/2}y)^2},$$

where x and y are positive.

A $\frac{10y^3\sqrt{y}}{x}$

B $\frac{50y^3}{x}$

C $\frac{50y^3\sqrt{y}}{x}$

D $\frac{50\sqrt{y}}{x}$

E $\frac{250y^3\sqrt{y}}{x}$

WORKED SOLUTION

$$(50x^{3/2}y^{-1})^2 = 2500x^3y^{-2} \text{ and } (2x^{-1}y^{5/2})^3 = 8x^{-3}y^{15/2}.$$

So the numerator is $2500x^3y^{-2} \cdot 8x^{-3}y^{15/2} = 20000y^{11/2}$.

The denominator is $(20x^{1/2}y)^2 = 400xy^2$.

Therefore

$$\frac{20000y^{11/2}}{400xy^2} = \frac{50y^{7/2}}{x} = \boxed{\frac{50y^3\sqrt{y}}{x}}.$$

M1-02

Successive percentage change and comparison

Answer B

QUESTION 2

The price of item P is reduced by 20%, and then a sales tax of 5% is applied to the reduced price.

The price of item Q is increased by 10%, and then reduced by 10%.

After these two-step changes, the final prices of P and Q are equal.

What was the ratio of the original price of P to the original price of Q?

A 28 : 33

B 33 : 28

C 11 : 10

D 6 : 5

E 99 : 80

WORKED SOLUTION

Final price of P is $0.8 \times 1.05P = 0.84P$.

Final price of Q is $1.1 \times 0.9Q = 0.99Q$.

So $0.84P = 0.99Q$, giving

$$\frac{P}{Q} = \frac{0.99}{0.84} = \frac{99}{84} = \frac{33}{28}.$$

M1-03

Algebraic fractions and hidden substitution

Answer D

QUESTION 3

For real x , define

$$E = \frac{\frac{1}{x-3} - \frac{1}{x+2}}{\frac{1}{x-3} + \frac{1}{x+2}}.$$

Given that $x > 3$ and

$$E + \frac{1}{E} = \frac{13}{2},$$

what is the value of x ?

A $\frac{11}{2}$

B 8

C 12

D $\frac{31}{2}$

E 18

WORKED SOLUTION

Simplify first:

$$E = \frac{\frac{(x+2)-(x-3)}{(x-3)(x+2)}}{\frac{(x+2)+(x-3)}{(x-3)(x+2)}} = \frac{5}{2x-1}.$$

Let $t = E$. Then $t + \frac{1}{t} = \frac{13}{2}$, so

$$2t^2 - 13t + 2 = 0 = (2t - 1)(t - 6).$$

Hence $t = 6$ or $t = \frac{1}{6}$. Since $x > 3$, we have $0 < E < 1$, so $E = \frac{1}{6}$.

Therefore

$$\frac{5}{2x-1} = \frac{1}{6} \Rightarrow 2x - 1 = 30 \Rightarrow \boxed{x = \frac{31}{2}}.$$

M1-04

Inequalities with domain restrictions

Answer A

QUESTION 4

How many integer values of x satisfy all three conditions?

- $-5 \leq x \leq 5$
- $\frac{x^2 + 3x - 10}{x^2 - 1}$ is defined
- $\frac{x^2 + 3x - 10}{x^2 - 1} \leq 1$

A 6

B 5

C 4

D 7

WORKED SOLUTION

Bring everything to one side:

$$\frac{x^2 + 3x - 10}{x^2 - 1} - 1 \leq 0 \Rightarrow \frac{3x - 9}{(x - 1)(x + 1)} \leq 0.$$

So we solve

$$\frac{x - 3}{(x - 1)(x + 1)} \leq 0, \quad x \neq \pm 1.$$

A sign chart gives $(-\infty, -1) \cup (1, 3]$.

Intersecting with $-5 \leq x \leq 5$ gives $-5, -4, -3, -2, 2, 3$, so there are 6 values.

M1-05

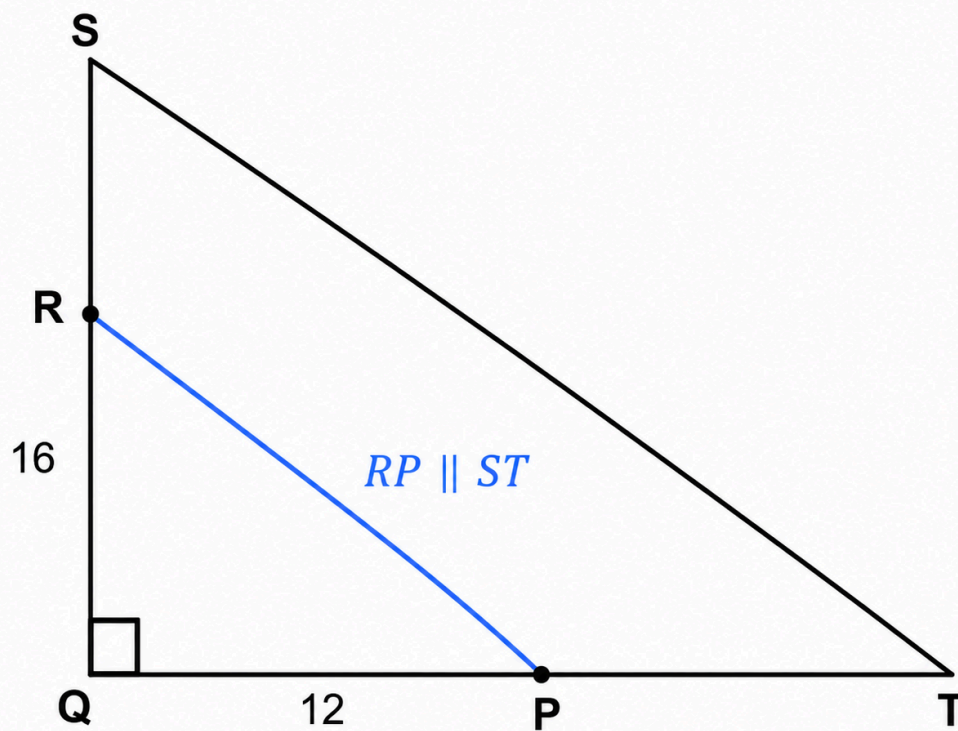
Similarity and area scaling in a right triangle

Answer E

QUESTION 5

Triangle SQT is right-angled at Q, with SQ = 16 cm and QT = 12 cm.

Point R lies on SQ such that SR : RQ = 1 : 3. Through R, a line parallel to ST meets QT at P.



What is the area of quadrilateral SRP T?

A 24 cm^2

B 30 cm^2

C 36 cm^2

D 40 cm^2

E 42 cm^2

WORKED SOLUTION

The area of the whole triangle is $\frac{1}{2} \cdot 16 \cdot 12 = 96$.

Since $SR : RQ = 1 : 3$, we get $RQ = \frac{3}{4} \cdot 16 = 12$.

Because $RP \parallel ST$, triangles QRP and QST are similar with scale factor

$$\frac{QR}{QS} = \frac{12}{16} = \frac{3}{4}.$$

So $QP = \frac{3}{4} \cdot 12 = 9$. Then area of triangle QRP is $\frac{1}{2} \cdot 12 \cdot 9 = 54$.

Hence quadrilateral $SRPT$ has area $96 - 54 = \boxed{42 \text{ cm}^2}$.

M1-06

Arithmetic sequences and common terms

Answer **C**

QUESTION 6

The n th term of sequence T is $4n - 3$.

The n th term of sequence V is $7n + p$, where p is an integer and $0 \leq p < 7$.

The sequences have consecutive common terms 53 and 81.

What is the value of p ?

A 1

B 2

C 4

D 5

E 6

WORKED SOLUTION

Using the common term 53,

$$53 = 7n + p.$$

Since $53 \equiv 4 \pmod{7}$, and $0 \leq p < 7$, we must have $p = 4$.

M1-07

Cylinders with swapped dimensions

Answer **B**

QUESTION 7

Two solid cylinders P and Q are defined using positive numbers x and y , where $x > y$.

- Cylinder P has diameter x and height y .
- Cylinder Q has diameter y and height x .

The total surface area of P is $\frac{8}{5}$ of the total surface area of Q.

The volume of P exceeds the volume of Q by $32\pi \text{ cm}^3$.

What is the value of $x + y$?

A 10

B 12

C 14

D 16

E 18

WORKED SOLUTION

For a cylinder of diameter d and height h , total surface area is

$$2\pi\left(\frac{d}{2}\right)^2 + 2\pi\left(\frac{d}{2}\right)h.$$

Thus

$$A_P = \pi \left(\frac{x^2}{2} + xy \right), \quad A_Q = \pi \left(\frac{y^2}{2} + xy \right).$$

Set $x = ky$. Then

$$\frac{k^2 + 2k}{1 + 2k} = \frac{8}{5}.$$

So $5k^2 - 6k - 8 = 0$, giving the positive root $k = 2$. Hence $x = 2y$.

Now

$$V_P - V_Q = \frac{\pi}{4}(x^2y - y^2x) = \frac{\pi}{4}xy(x - y) = 32\pi.$$

Substitute $x = 2y$:

$$\frac{\pi}{4}(2y)(y)(y) = 32\pi \Rightarrow \frac{y^3}{2} = 32 \Rightarrow y^3 = 64 \Rightarrow y = 4.$$

Hence $x = 8$, so $x + y = \boxed{12}$.

M1-08

Ratio change after equal spending

Answer **D**

QUESTION 8

Pat and Alex have a combined total of £ 84. The ratio of Pat's money to Alex's money is 5 : 2.

They each spend the same amount on sweets.

After this, Pat receives £ 6 and Alex receives £ 1. The ratio of their new amounts is then 3 : 1.

How much did each of them spend?

A £ 3

B £ 4

C £ 4.25

D £ 4.50

E £ 6

WORKED SOLUTION

Initially the amounts are 60 and 24.

If each spends s , then afterwards they have $60 - s$ and $24 - s$. After receiving the extra money, they have $66 - s$ and $25 - s$.

So

$$66 - s = 3(25 - s) \Rightarrow 66 - s = 75 - 3s \Rightarrow 2s = 9 \Rightarrow s = 4.5.$$

Each spent £ 4.50.

M1-09

Coordinate geometry of a tangent

Answer A

QUESTION 9

The circle $x^2 + y^2 = 25$ has centre at the origin.

A tangent to the circle passes through the point $(1, 7)$ and has positive gradient.

What is the equation of the tangent?

A $3x - 4y + 25 = 0$

B $4x - 3y + 25 = 0$

C $3x + 4y - 25 = 0$

D $4x + 3y - 25 = 0$

E

$$y = \frac{4}{3}x + \frac{17}{3}$$

WORKED SOLUTION

Let the tangent be $y = mx + c$. Since it passes through $(1, 7)$, we have $c = 7 - m$.

The distance from the origin to the tangent must equal the radius 5:

$$\frac{|c|}{\sqrt{m^2 + 1}} = 5.$$

So

$$(7 - m)^2 = 25(m^2 + 1) \Rightarrow 12m^2 + 7m - 12 = 0 = (4m - 3)(3m + 4).$$

The positive gradient is $m = \frac{3}{4}$. Then $c = 7 - \frac{3}{4} = \frac{25}{4}$.

Hence

$$y = \frac{3}{4}x + \frac{25}{4} \Leftrightarrow \boxed{3x - 4y + 25 = 0}.$$

M1-10

Conditional probability with non-standard dice

Answer E**QUESTION 10**

Two fair dice X and Y are available.

- The faces on X are labelled 1, 1, 2, 3, 4, 5.
- The faces on Y are labelled 2, 3, 3, 4, 5, 6.

One of the dice is chosen at random and rolled twice.

Given that the total shown is 7, what is the probability that the chosen die was X ?

A

$$\frac{1}{3}$$

B $\frac{3}{8}$

C $\frac{1}{2}$

D $\frac{3}{5}$

E $\frac{2}{5}$

WORKED SOLUTION

For die X, the ordered outcomes summing to 7 are (2, 5), (5, 2), (3, 4), (4, 3): 4 outcomes out of 36. So $P(7 | X) = \frac{1}{9}$.

For die Y, 3 appears twice, so the ordered outcomes summing to 7 are 6 in total. Hence $P(7 | Y) = \frac{1}{6}$.

With equal prior probability of choosing each die,

$$P(X | 7) = \frac{\frac{1}{2} \cdot \frac{1}{9}}{\frac{1}{2} \cdot \frac{1}{9} + \frac{1}{2} \cdot \frac{1}{6}} = \frac{\frac{1}{9}}{\frac{1}{9} + \frac{1}{6}} = \frac{2}{5}.$$

M1-11

Exponential equation via substitution

Answer **C**

QUESTION 11

The equation $2^x + 2^{2-x} = 5$ has two real solutions.

What is the larger possible value of x?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Let $t = 2^x$. Then $2^{2-x} = \frac{4}{t}$, so

$$t + \frac{4}{t} = 5 \Rightarrow t^2 - 5t + 4 = 0 = (t - 1)(t - 4).$$

Thus $t = 1$ or $t = 4$, so $x = 0$ or $x = 2$. The larger value is 2.

M1-12

Rearranging to make a variable the subject

Answer B

QUESTION 12

Suppose x and y satisfy

$$x = y + \frac{4}{y},$$

where $y > 0$ and $x > 4$.

Which of the following is the larger possible value of y in terms of x ?

A $\frac{x - \sqrt{x^2 - 16}}{2}$

B $\frac{x + \sqrt{x^2 - 16}}{2}$

C $\frac{x + \sqrt{x^2 + 16}}{2}$

D $\frac{2}{x + \sqrt{x^2 - 16}}$

E $\frac{4}{x + \sqrt{x^2 - 16}}$

WORKED SOLUTION

Multiply by y :

$$xy = y^2 + 4 \Rightarrow y^2 - xy + 4 = 0.$$

Using the quadratic formula,

$$y = \frac{x \pm \sqrt{x^2 - 16}}{2}.$$

The larger possible value uses the positive sign, so

$$y = \frac{x + \sqrt{x^2 - 16}}{2}.$$

M1-13

Surface area comparison of a cylinder and a cube

Answer D

QUESTION 13

A solid cylinder has radius r cm and height h cm.

A cube has side length $\sqrt{\pi} r$ cm.

The total surface area of the cylinder is equal to four times the total surface area of the cube.

What is $\frac{h}{r}$?

A 7

B 9

C 10

D 11

E 12

WORKED SOLUTION

The cylinder's total surface area is $2\pi r^2 + 2\pi rh = 2\pi r(r + h)$.

The cube's total surface area is $6(\sqrt{\pi}r)^2 = 6\pi r^2$.

Given the cylinder equals four times the cube,

$$2\pi r(r + h) = 24\pi r^2.$$

Cancel $2\pi r$:

$$r + h = 12r \Rightarrow h = 11r.$$

Therefore $\frac{h}{r} = \boxed{11}$.

M1-14

Mean and median with an ordered list

Answer A

QUESTION 14

Consider the list

$$1, 2, x, x + 1, 2x - 1, 15,$$

where $2 < x < 8$.

The mean of the six numbers is one more than the median.

What is the value of x ?

A $\frac{9}{2}$

B 4

C 5

D $\frac{11}{2}$

E 6

WORKED SOLUTION

Since $x > 2$, the list is already ordered. So the median is

$$\frac{x + (x + 1)}{2} = x + \frac{1}{2}.$$

The mean is one more than this, so the mean is $x + \frac{3}{2}$.

The mean is also

$$\frac{1 + 2 + x + (x + 1) + (2x - 1) + 15}{6} = \frac{4x + 18}{6}.$$

Therefore

$$\frac{4x + 18}{6} = x + \frac{3}{2} \Rightarrow 4x + 18 = 6x + 9 \Rightarrow 2x = 9 \Rightarrow \boxed{x = \frac{9}{2}}.$$

QUESTION 15

A cyclist rides from A to B uphill at 12 km h^{-1} , immediately turns round, and returns from B to A along the same route at 18 km h^{-1} .

She spends exactly 6 minutes stationary at B.

The total time for the complete trip is 46 minutes.

What is the distance from A to B?

A 3.6 km

B 4.0 km

C 4.2 km

D 4.5 km

E 4.8 km

WORKED SOLUTION

Excluding the 6-minute stop, the travel time is 40 minutes = $\frac{2}{3}$ hours.

If the one-way distance is d km, then

$$\frac{d}{12} + \frac{d}{18} = \frac{2}{3}.$$

Since $\frac{1}{12} + \frac{1}{18} = \frac{5}{36},$

$$d \cdot \frac{5}{36} = \frac{2}{3} \Rightarrow d = \frac{2}{3} \cdot \frac{36}{5} = \frac{24}{5} = 4.8.$$

So the distance is 4.8 km.

M1-16

Exact trigonometric values

Answer **B**

QUESTION 16

Given that

$$y = \frac{2 \sin 15^\circ}{\cos 30^\circ},$$

what is the value of y^2 ?

A $\frac{4 - 2\sqrt{3}}{3}$

B $\frac{8 - 4\sqrt{3}}{3}$

C $\frac{8 + 4\sqrt{3}}{3}$

D $2 - \sqrt{3}$

E $4 - 2\sqrt{3}$

WORKED SOLUTION

Use

$$\sin 15^\circ = \sin(45^\circ - 30^\circ) = \frac{\sqrt{6} - \sqrt{2}}{4}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}.$$

Hence

$$y = \frac{2 \cdot \frac{\sqrt{6}-\sqrt{2}}{4}}{\frac{\sqrt{3}}{2}} = \frac{\sqrt{6}-\sqrt{2}}{\sqrt{3}}.$$

Therefore

$$y^2 = \frac{(\sqrt{6}-\sqrt{2})^2}{3} = \frac{6+2-2\sqrt{12}}{3} = \frac{8-4\sqrt{3}}{3}.$$

M1-17

Inverse proportion in standard form

Answer C

QUESTION 17

The variable t is inversely proportional to the square of w .

When $w = 2 \times 10^3$, the value of t is 4.5×10^{-5} .

What is the value of w when $t = 8 \times 10^{-6}$?

A 7.5×10^3

B $3\sqrt{10} \times 10^2$

C $1.5\sqrt{10} \times 10^3$

D $2\sqrt{10} \times 10^3$

E 4.5×10^3

WORKED SOLUTION

Since $t = \frac{k}{w^2}$, use the first condition:

$$k = (4.5 \times 10^{-5})(2 \times 10^3)^2 = (4.5 \times 10^{-5})(4 \times 10^6) = 180.$$

Now solve

$$8 \times 10^{-6} = \frac{180}{w^2} \Rightarrow w^2 = 2.25 \times 10^7.$$

So

$$w = \sqrt{2.25 \times 10^7} = 1.5\sqrt{10} \times 10^3.$$

M1-18

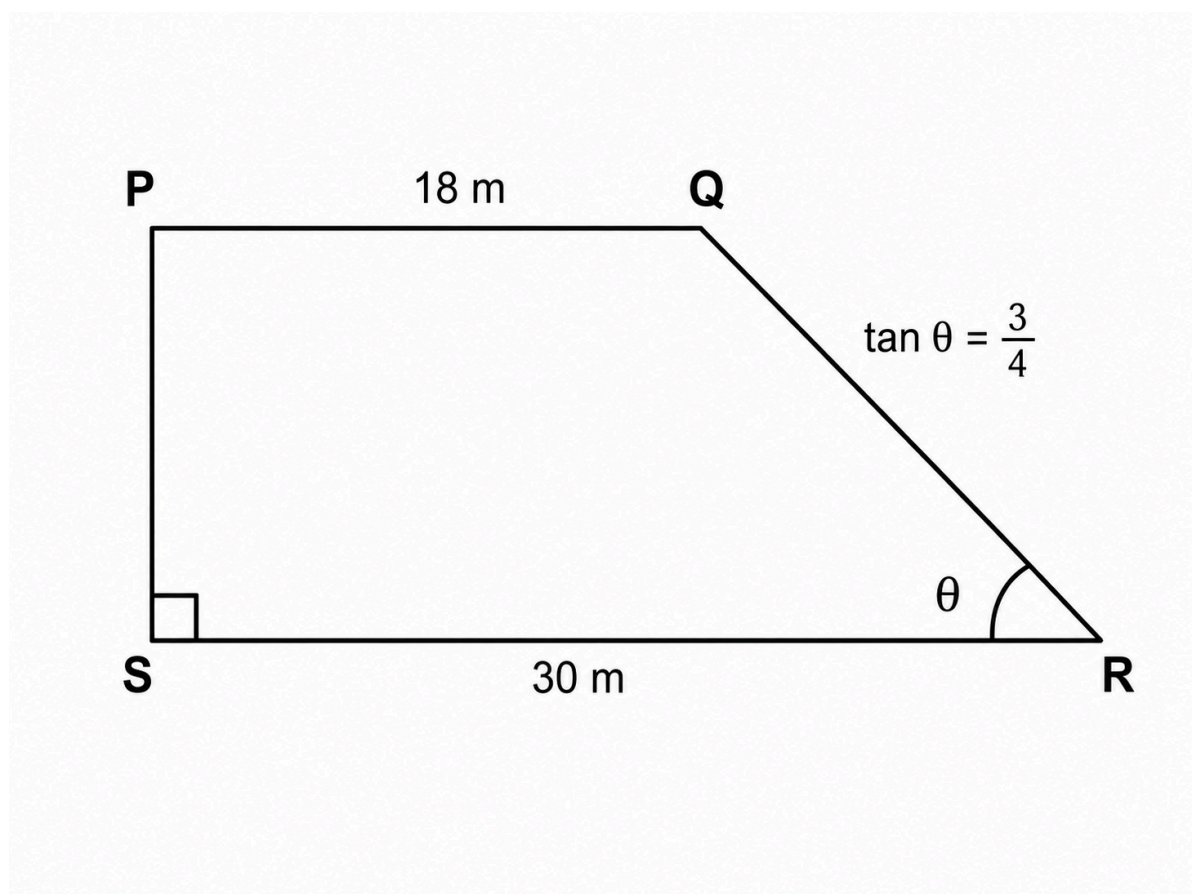
Trapezium area using trigonometry

Answer D

QUESTION 18

PQRS is a trapezium in which $PQ \parallel SR$ and $PS \perp SR$.

Also, $PQ = 18 \text{ m}$, $SR = 30 \text{ m}$, and $\tan \angle QRS = \frac{3}{4}$.



What is the area of the trapezium?

A 180 m^2

B 192 m^2

C 204 m^2

D 216 m^2

E 240 m^2

WORKED SOLUTION

The difference between the parallel sides is $30 - 18 = 12$, which is the horizontal run of the sloping side.

If the height is h , then

$$\tan \angle QRS = \frac{h}{12} = \frac{3}{4} \Rightarrow h = 9.$$

So the area is

$$\frac{1}{2}(18 + 30)(9) = 24 \cdot 9 = \boxed{216 \text{ m}^2}.$$

M1-19

Quadratic roots and discriminant reasoning

Answer A

QUESTION 19

The equation $x^2 + bx + 8 = 0$ has two real roots whose positive difference is 2. The sum of the roots is negative.

What is the value of b ?

A 6

B 5

C 4

D -6

E -4

WORKED SOLUTION

If the roots are r and s , then $(r - s)^2 = (r + s)^2 - 4rs$.

Here $r + s = -b$ and $rs = 8$. Since the difference is 2,

$$4 = b^2 - 32 \Rightarrow b^2 = 36.$$

So $b = \pm 6$. The sum of the roots is negative, so $-b < 0 \Rightarrow b > 0$. Hence

$$\boxed{b = 6}.$$

M1-20

Algebraic geometry with area change

Answer **E**

QUESTION 20

A rectangle has length $(2x - 1)$ cm and width $(x + 1)$ cm.

A larger rectangle is made by adding 3 cm to both the length and the width.

The larger rectangle has twice the area of the original rectangle.

What is the perimeter of the original rectangle?

A 24 cm

B 26 cm

C 28 cm

D 29 cm

E 30 cm

WORKED SOLUTION

Original area is $(2x - 1)(x + 1)$. New dimensions are $2x + 2$ and $x + 4$, so the new area is $(2x + 2)(x + 4)$.

Given new area is twice old area,

$$(2x + 2)(x + 4) = 2(2x - 1)(x + 1).$$

This simplifies to

$$x^2 - 4x - 5 = 0 = (x - 5)(x + 1).$$

So $x = 5$. The original dimensions are 9 and 6, so the perimeter is

$$2(9 + 6) = \boxed{30 \text{ cm}}.$$

M1-21

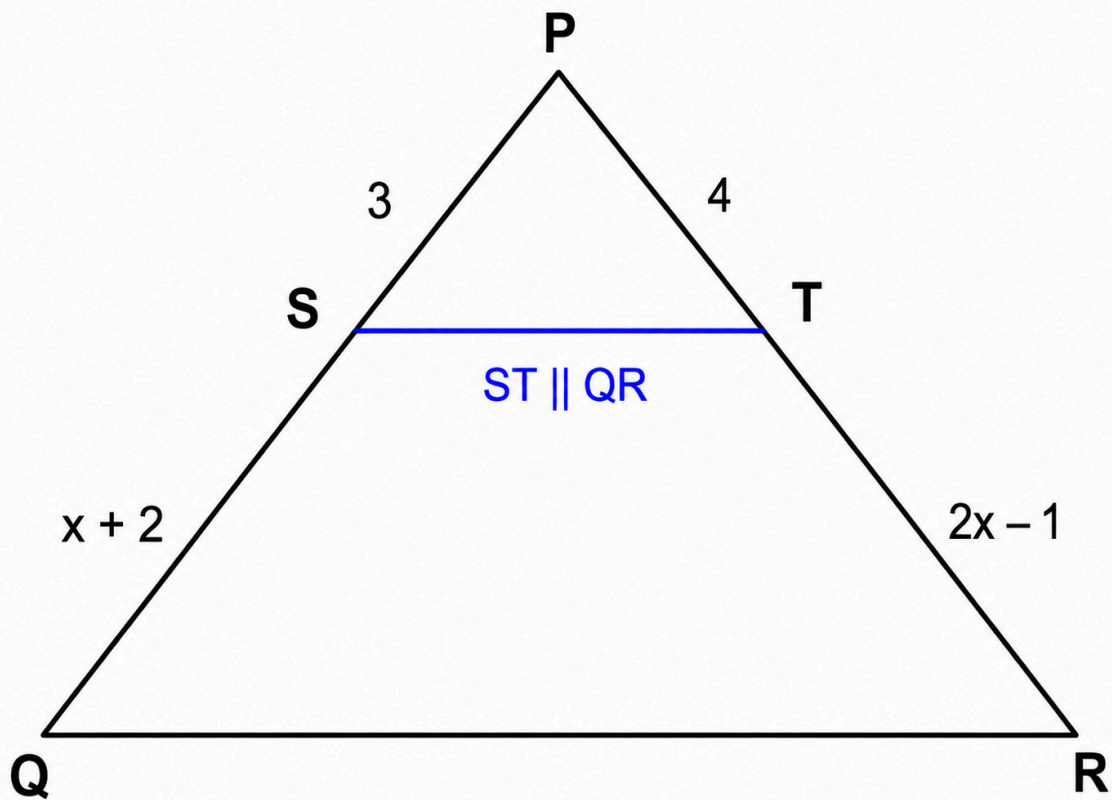
Similarity and area scaling

Answer **C**

QUESTION 21

In triangle PQR , point S lies on PQ and point T lies on PR . The line ST is parallel to QR .

The lengths are labelled as follows: $PS = 3 \text{ cm}$, $SQ = (x + 2) \text{ cm}$, $PT = 4 \text{ cm}$, $TR = (2x - 1) \text{ cm}$.



Given that the area of triangle PST is 12 cm^2 , what is the area of trapezium $STQR$?

A 96 cm^2

B 120 cm^2

C 135 cm^2

D 144 cm^2

E 147 cm^2

WORKED SOLUTION

Since $ST \parallel QR$, triangles PST and PQR are similar, so

$$\frac{PS}{PQ} = \frac{PT}{PR}.$$

Now $PQ = 3 + (x + 2) = x + 5$ and $PR = 4 + (2x - 1) = 2x + 3$.

Therefore

$$\frac{3}{x + 5} = \frac{4}{2x + 3}.$$

So $6x + 9 = 4x + 20$, giving $x = \frac{11}{2}$.

Then

$$\frac{PS}{PQ} = \frac{3}{21/2} = \frac{2}{7}.$$

Hence the area scale factor is $\left(\frac{2}{7}\right)^2 = \frac{4}{49}$.

So if area $PST = 12$, then area $PQR = 12 \cdot \frac{49}{4} = 147$.

Thus the area of trapezium $STQR$ is $147 - 12 = \boxed{135 \text{ cm}^2}$.

M1-22

Probability with two colours

Answer **B**

QUESTION 22

Charlie has a bowl containing red sweets and green sweets only. There are 9 more red sweets than green sweets.

Two sweets are chosen at random without replacement.

The probability that the two sweets are the same colour is $\frac{4}{7}$.

How many sweets are there in the bowl altogether?

A 18

B 21

C 24

D 27

E 30

WORKED SOLUTION

Let the number of green sweets be g . Then the number of red sweets is $g + 9$, so total sweets are $2g + 9$.

The probability condition is

$$\frac{\binom{g+9}{2} + \binom{g}{2}}{\binom{2g+9}{2}} = \frac{4}{7}.$$

This simplifies to

$$(g - 6)(g + 18) = 0.$$

So $g = 6$. Then red sweets = 15, giving total sweets = 21.

M1-23

Successive translations and vector equations

Answer **D**

QUESTION 23

The point $(-1, 5)$ is translated by two successive vectors.

The first translation is by $\begin{pmatrix} 3p \\ 2 - q \end{pmatrix}$, and the second is by $\begin{pmatrix} q + 1 \\ p - 3 \end{pmatrix}$.

The final image is $(8, 4)$.

What is the value of $p^2 + q^2$?

A 4

B 5

C 6

D 8

E 10

WORKED SOLUTION

The total translation vector is

$$\begin{pmatrix} 3p + q + 1 \\ p - q - 1 \end{pmatrix}.$$

Since $(-1, 5) \rightarrow (8, 4)$, the net translation is $\begin{pmatrix} 9 \\ -1 \end{pmatrix}$.

So

$$3p + q = 8, \quad p - q = 0.$$

Thus $p = q$, so $4p = 8 \Rightarrow p = 2$, hence $q = 2$.

Therefore $p^2 + q^2 = 4 + 4 = \boxed{8}$.

M1-24

Quadratic graphs and minimum values

Answer **A**

QUESTION 24

Consider the family of graphs

$$y = ax^2 + 2(a - 1)x + a + 3.$$

What is the complete range of values of a for which the minimum point of the graph lies above the x -axis?

A $a > \frac{1}{5}$

B $a > 0$

C $a \geq \frac{1}{5}$

D $a < 0$

E $0 < a < \frac{1}{5}$

WORKED SOLUTION

For there to be a minimum, we need $a > 0$.

For $y = ax^2 + bx + c$, the minimum value is $c - \frac{b^2}{4a}$.

Here $b = 2(a - 1)$ and $c = a + 3$, so the minimum value is

$$a + 3 - \frac{(2(a - 1))^2}{4a} = a + 3 - \frac{(a - 1)^2}{a} = 5 - \frac{1}{a}.$$

We need this to be positive:

$$5 - \frac{1}{a} > 0.$$

Since $a > 0$, this gives $5a > 1 \Rightarrow a > \frac{1}{5}$.

M1-25

Counting constrained selections

Answer **E**

QUESTION 25

An online company sells storage containers. The following items are available:

Capacity of container	Number available
2 litres	2
3 litres	3

Capacity of container	Number available
5 litres	4

A customer chooses a selection of containers with total capacity 23 litres.

Different selections are distinguished only by how many containers of each size are chosen.

If all possible selections are equally likely, what is the probability that the chosen selection contains at least one 2-litre container?

A $\frac{1}{3}$

B $\frac{1}{2}$

C $\frac{3}{5}$

D $\frac{3}{4}$

E $\frac{2}{3}$

WORKED SOLUTION

Let a , b , c be the numbers of 2-, 3- and 5-litre containers. Then

$$2a + 3b + 5c = 23,$$

with $0 \leq a \leq 2$, $0 \leq b \leq 3$, $0 \leq c \leq 4$.

The valid selections are

- (0, 1, 4)
- (1, 2, 3)
- (2, 3, 2)

Two of the three contain at least one 2-litre container, so the probability is $\frac{2}{3}$.

M1-26

Similarity and three-dimensional scaling

Answer C

QUESTION 26

The space diagonal of a cuboid is $7\sqrt{13}$ cm.

A similar cuboid has volume 8 times the volume of the original cuboid.

What is the space diagonal of the larger cuboid?

A $7\sqrt{26}$ cm

B $14\sqrt{7}$ cm

C $14\sqrt{13}$ cm

D $21\sqrt{13}$ cm

E $28\sqrt{13}$ cm

WORKED SOLUTION

A volume scale factor of 8 means a linear scale factor of $\sqrt[3]{8} = 2$.

So the new diagonal is

$$2 \cdot 7\sqrt{13} = \boxed{14\sqrt{13} \text{ cm}}.$$

QUESTION 27

Alex, Cameron and Sam are running a 400 m race. They each run at a different constant speed.

When Alex finishes, Cameron has run 300 m and Sam has run 240 m.

Alex beats Sam by $33\frac{1}{3}$ seconds.

What is Cameron's speed?

A 5 m s^{-1}

B 6 m s^{-1}

C 6.4 m s^{-1}

D 7 m s^{-1}

E 8 m s^{-1}

WORKED SOLUTION

Let Alex's speed be A , Cameron's speed be C , and Sam's speed be S .

When Alex finishes, the elapsed time is $\frac{400}{A}$. In that time, Cameron runs 300 m, so

$$C \cdot \frac{400}{A} = 300 \Rightarrow \frac{C}{A} = \frac{3}{4}.$$

Similarly, Sam runs 240 m, so

$$S \cdot \frac{400}{A} = 240 \Rightarrow \frac{S}{A} = \frac{3}{5}.$$

$$\text{Thus } S = \frac{3}{5}A.$$

Alex beats Sam by $\frac{100}{3}$ s, so

$$\frac{400}{S} - \frac{400}{A} = \frac{100}{3}.$$

Substitute $S = \frac{3}{5}A$:

$$\frac{400}{3A/5} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{2000}{3A} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{800}{3A} = \frac{100}{3}.$$

So $A = 8 \text{ m s}^{-1}$. Hence

$$C = \frac{3}{4}A = 6 \text{ m s}^{-1}.$$

Therefore Cameron's speed is 6 m s^{-1} .

Mathematics 2

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 5 · preserved Mathematics 2 module. Question wording, option order, answer and solution method are unchanged.

M2-01

Polynomials

Answer **D****QUESTION 1**

The polynomial $P(x) = x^4 + ax^3 + bx^2 - 44x + 24$ has $(x - 1)$ and $(x - 2)$ as factors. What is the value of $a^2 + b$?

A 49**B** 61**C** 68**D** 75**E** 82**WORKED SOLUTION**

Since $x = 1$ and $x = 2$ are roots, $P(1)=0$ and $P(2)=0$.

From $P(1)=0$: $1 + a + b - 44 + 24 = 0$, so $a + b = 19$.

From $P(2)=0$: $16 + 8a + 4b - 88 + 24 = 0$, so $2a + b = 12$.

Subtracting gives $a = -7$, and then $b = 26$.

Hence $a^2 + b = 49 + 26 = \mathbf{75}$.

M2-02

Calculus / stationary points

Answer **B****QUESTION 2**

The curve $y = x^3 + px^2 + qx + 10$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. What is the sum of the y-coordinates of these two stationary points?

A -8**B** -2**C** 2**D** 6**E** 12**WORKED SOLUTION**

Differentiate: $y' = 3x^2 + 2px + q$. Since the stationary points are at $x = -1$ and $x = 3$, we have $y' = 3(x+1)(x-3) = 3x^2 - 6x - 9$.

So $2p = -6$ and $q = -9$, hence $p = -3$.

The curve is $y = x^3 - 3x^2 - 9x + 10$.

At $x = -1$, $y = -1 - 3 + 9 + 10 = 15$.

At $x = 3$, $y = 27 - 27 - 27 + 10 = -17$.

The sum is $15 + (-17) = -2$.

M2-03

Coordinate geometry

Answer **E****QUESTION 3**

The line $y = mx + 3$ is reflected in the line $y = x$. The reflected line has y-intercept $-\frac{2}{3}$. What is the value of m ?

A $\frac{3}{2}$

B 2

C $\frac{5}{2}$

D 3

E $\frac{9}{2}$

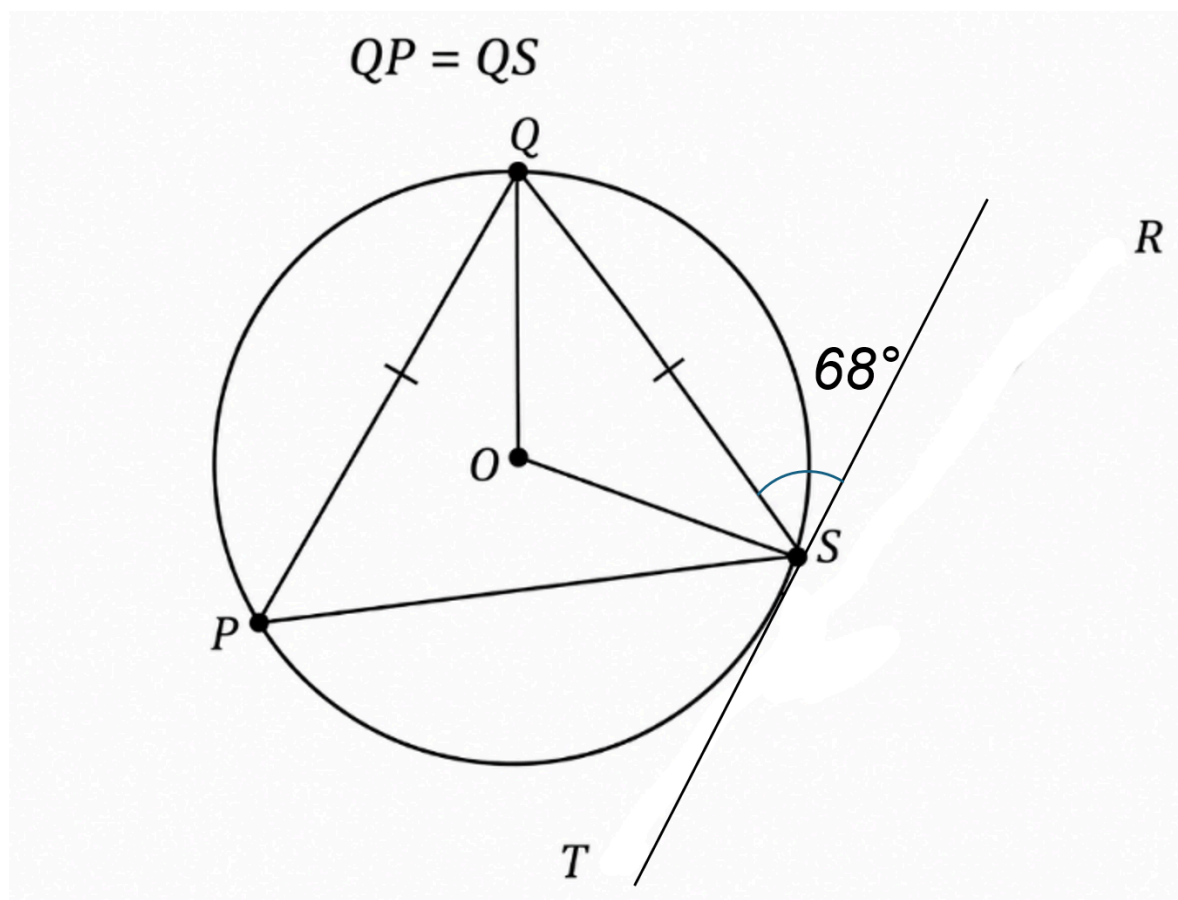
WORKED SOLUTION

Reflecting in $y = x$ swaps x and y . So the reflected line is $x = my + 3$, which rearranges to $y = (x - 3)/m = (1/m)x - 3/m$.

The y -intercept of the reflected line is therefore $-3/m$.

Given $-3/m = -2/3$, we get $3/m = 2/3$, so $9 = 2m$ and $m = \mathbf{9/2}$.

QUESTION 4



In the diagram, O is the centre of the circle and RT is a tangent at S . Points P and Q lie on the circumference, $QP = QS$, and the marked angle is $\angle QSR = 68^\circ$. What is the size of $\angle POS$?

A 44°

B 68°

C 88°

D 112°

E 136°

WORKED SOLUTION

By the tangent–chord theorem applied to chord QS,

$$\angle QPS = \angle QSR = 68^\circ.$$

Since $QP = QS$, triangle PQS is isosceles with base PS. Hence

$$\angle PSQ = 68^\circ, \quad \angle PQS = 180^\circ - 68^\circ - 68^\circ = 44^\circ.$$

Chord PS subtends angle $\angle PQS$ at the circumference and angle $\angle POS$ at the centre. Therefore

$$\angle POS = 2\angle PQS = 2(44^\circ) = \boxed{88^\circ}.$$

M2-05

Integration / area

Answer **B**

QUESTION 5

What is the total area enclosed between the curve $y = x^2 - 4x + 3$ and the line $y = -x + 1$, for $0 \leq x \leq 3$?

A $\frac{7}{6}$

B $\frac{11}{6}$

C $\frac{13}{6}$

D $\frac{17}{6}$

E $\frac{23}{6}$

WORKED SOLUTION

Subtract the line from the curve: $(x^2 - 4x + 3) - (-x + 1) = x^2 - 3x + 2 = (x-1)(x-2)$.

This is positive on $[0, 1]$, negative on $[1, 2]$, and positive on $[2, 3]$.

So total area = $\int_0^1 (x^2 - 3x + 2) dx - \int_1^2 (x^2 - 3x + 2) dx + \int_2^3 (x^2 - 3x + 2) dx$.

An antiderivative is $x^3/3 - 3x^2/2 + 2x$.

Evaluating gives $5/6 + 1/6 + 5/6 = 11/6$.

M2-06

Surds

Answer C

QUESTION 6

Simplify $\sqrt{(18 + 8\sqrt{5})} - \sqrt{(18 - 8\sqrt{5})}$. Write the answer in the form $k\sqrt{2}$. What is k ?

A 2

B 3

C 4

D 5

E 6

WORKED SOLUTION

Write $18 + 8\sqrt{5}$ as $a + b + 2\sqrt{ab}$ with $a + b = 18$ and $ab = 80$. The numbers are 10 and 8.

So $\sqrt{(18 + 8\sqrt{5})} = \sqrt{10 + \sqrt{8}} = \sqrt{10 + 2\sqrt{2}}$.

Similarly, $\sqrt{(18 - 8\sqrt{5})} = \sqrt{10 - 2\sqrt{2}}$.

Subtracting gives $(\sqrt{10 + 2\sqrt{2}}) - (\sqrt{10 - 2\sqrt{2}}) = 4\sqrt{2}$.

Hence $k = 4$.

M2-07

Tangents

Answer B

QUESTION 7

The line $y = 2x + k$ is a tangent to the curve $y = 3x^2 - 10x + 7$. What is the value of k ?

A -7**B** -5**C** -3**D** 1**E** 5**WORKED SOLUTION**

At the point of tangency the line and curve have one repeated intersection.

Set them equal: $3x^2 - 10x + 7 = 2x + k$, so $3x^2 - 12x + (7 - k) = 0$.

For tangency the discriminant is 0:

$$(-12)^2 - 4 \cdot 3 \cdot (7 - k) = 0.$$

$$144 - 12(7 - k) = 0 \Rightarrow 144 - 84 + 12k = 0 \Rightarrow 60 + 12k = 0.$$

So $k = -5$.

M2-08

Differentiation

Answer B

QUESTION 8

Given $y = (3\sqrt{x} - 2/\sqrt{x})^2$, find dy/dx when $x = \frac{1}{2}$.

A -9

B -7

C -5

D 7

E 9

WORKED SOLUTION

First simplify: $y = 9x - 12 + 4/x$.

Differentiate: $dy/dx = 9 - 4/x^2$.

At $x = 1/2$, $x^2 = 1/4$, so $4/x^2 = 16$.

Therefore $dy/dx = 9 - 16 = -7$.

M2-09

Integration

Answer **C**

QUESTION 9

Find the value of $\int_1^4 (3x^2 + 1)/(x\sqrt{x}) \, dx$.

A 12

B 14

C 15

D 16

E 18

WORKED SOLUTION

Rewrite the integrand: $(3x^2 + 1)/(x\sqrt{x}) = 3x^{1/2} + x^{-3/2}$.

So $\int (3x^{1/2} + x^{-3/2}) dx = 2x^{3/2} - 2x^{-1/2}$.

Evaluate from 1 to 4:

$$[2x^{3/2} - 2x^{-1/2}]_1^4 = (2 \cdot 8 - 2 \cdot 1/2) - (2 \cdot 1 - 2 \cdot 1) = 15 - 0 = 15.$$

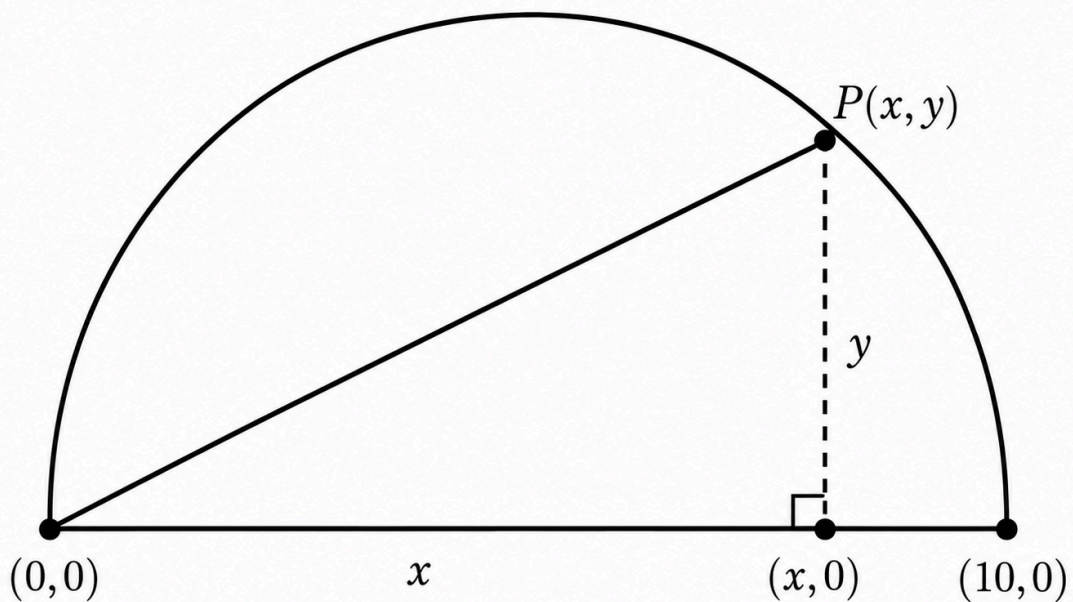
M2-10

Coordinate geometry / optimisation

Answer **B**

QUESTION 10

P lies on the semicircle of radius 5



A point $P(x,y)$ lies on the semicircle with diameter from $(0,0)$ to $(10,0)$. A right triangle has vertices at $(0,0)$, $(x,0)$, and $P(x,y)$. What is the maximum possible area of the triangle?

A $25\sqrt{3/8}$

B $75\sqrt{3/8}$

C $25\sqrt[3]{4}$

D $75\sqrt[3]{4}$

E $125\sqrt[3]{8}$

WORKED SOLUTION

The semicircle has centre $(5,0)$ and radius 5, so $y = \sqrt{(25 - (x-5)^2)} = \sqrt{(10x - x^2)}$.

The triangle area is $A = \frac{1}{2}xy = \frac{1}{2}x\sqrt{(10x - x^2)}$.

To maximise A , maximise $A^2 = \frac{1}{4}x^2(10x - x^2) = \frac{1}{4}(10x^3 - x^4)$.

Differentiate: $d(A^2)/dx = \frac{1}{4}(30x^2 - 4x^3) = x^2(15 - 2x)/2$.

So the maximum occurs at $x = 15/2$.

Then $y = \sqrt{(10 \cdot 15/2 - (15/2)^2)} = \sqrt{(75/4)} = 5\sqrt{3}/2$.

Hence $A = \frac{1}{2} \cdot 15/2 \cdot 5\sqrt{3}/2 = 75\sqrt{3}/8$.

M2-11

Inequalities

Answer **B**

QUESTION 11

What is the complete set of real values of x for which $x^2(x^2 + 4) < 21$?

A $-2 < x < 2$

B $-\sqrt{3} < x < \sqrt{3}$

C $x < -\sqrt{3}$ or $x > \sqrt{3}$

D $-\sqrt{7} < x < \sqrt{7}$

E $-1 < x < 1$

WORKED SOLUTION

Let $u = x^2$, so $u \geq 0$. Then the inequality becomes $u(u+4) < 21$, i.e. $u^2 + 4u - 21 < 0$.

Factor: $(u+7)(u-3) < 0$, so $-7 < u < 3$.

Since $u = x^2 \geq 0$, we get $0 \leq u < 3$.

Thus $x^2 < 3$, so $-\sqrt{3} < x < \sqrt{3}$.

M2-12

Similar solids

Answer B

QUESTION 12

Four mathematically similar solids W, X, Y and Z satisfy the following:

- the ratio of lengths W:X is 1:2
- the ratio of surface areas X:Y is 4:9
- the ratio of volumes Y:Z is 8:27

What is the ratio of the volumes of W:Z?

A 1:81**B** 8:729**C** 4:243**D** 8:243**E** 16:729**WORKED SOLUTION**

Length ratio W:X = 1:2.

Surface area ratio X:Y = 4:9, so length ratio X:Y = 2:3.

Volume ratio Y:Z = 8:27, so length ratio Y:Z = 2:3.

Hence the overall length ratio $W:Z = 1:2 \times 2:3 \times 2:3 = 2:9$.

Volumes scale as the cube of length, so $W:Z$ volume ratio = $2^3:9^3 = 8:729$.

M2-13

Logarithms

Answer C

QUESTION 13

What is the mean of $\log_2 64$, $\log_4 64$, and $\log_8 64$?

A 3

B $10/3$

C $11/3$

D 4

E $13/3$

WORKED SOLUTION

$\log_2 64 = 6$, $\log_4 64 = 3$, and $\log_8 64 = 2$.

So the mean is $(6 + 3 + 2)/3 = 11/3$.

M2-14

Trigonometric comparison

Answer B

QUESTION 14

Which of the following is largest in value? (All angles are in radians.)

A $\sin 1$

B $\cos 0.5$

C $\tan 0.5$

D $\sin 0.5$

E $\cos 1$

WORKED SOLUTION

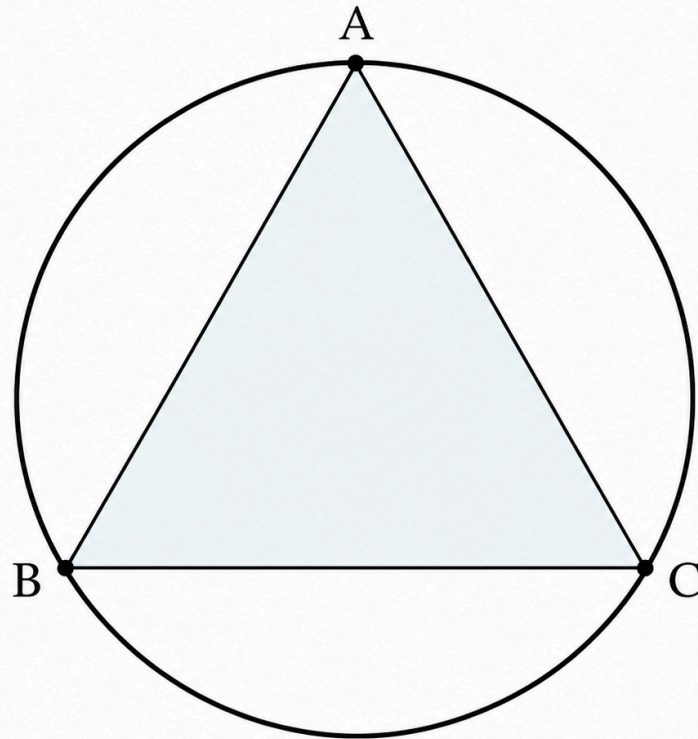
On $0 < \theta < \pi/2$, $\sin \theta$ and $\tan \theta$ increase, while $\cos \theta$ decreases.

Now $\sin 1 = \cos(\pi/2 - 1) = \cos(0.5708\dots)$. Since $0.5 < 0.5708\dots$, and cosine decreases on this interval, $\cos 0.5 > \cos 0.5708\dots = \sin 1$.

Also $\tan 0.5 < 1$ and in fact $\tan 0.5 < \cos 0.5$ numerically, while $\sin 0.5$ and $\cos 1$ are smaller still.

So the largest value is **$\cos 0.5$** .

QUESTION 15



$$AB = BC = CA = 6$$

An equilateral triangle of side 6 cm is inscribed in a circle. What is the area inside the circle but outside the triangle?

A $12\pi - 6\sqrt{3}$

B $12\pi - 9\sqrt{3}$

C $6\pi - 9\sqrt{3}$

D $4\pi - 9\sqrt{3}$

E $12\pi - 18\sqrt{3}$

WORKED SOLUTION

For an equilateral triangle of side s , the circumradius is $R = s/\sqrt{3}$.

So here $R = 6/\sqrt{3} = 2\sqrt{3}$.

Circle area $= \pi R^2 = \pi(2\sqrt{3})^2 = 12\pi$.

Triangle area $= (\sqrt{3}/4)s^2 = (\sqrt{3}/4) \cdot 36 = 9\sqrt{3}$.

Required area $= 12\pi - 9\sqrt{3}$, so the answer is **$12\pi - 9\sqrt{3}$** .

M2-16

Sequences and series

Answer C

QUESTION 16

An arithmetic sequence has first term a and common difference a . If S_n is the sum of the first n terms, which of the following is equal to $S_{2n} - 3S_n$?

A $an(n+1)/2$

B $an(2n-1)/2$

C $an(n-1)/2$

D $a(n-1)(n+1)$

E an^2

WORKED SOLUTION

The terms are $a, 2a, 3a, \dots, na$, so $S_n = a(1 + 2 + \dots + n) = an(n+1)/2$.

Then $S_{2n} = a \cdot 2n(2n+1)/2 = an(2n+1)$.

So $S_{2n} - 3S_n = an(2n+1) - 3[an(n+1)/2]$

$= [2an(2n+1) - 3an(n+1)]/2 = an(n-1)/2$.

Thus the correct expression is **$an(n-1)/2$** .

M2-17

Exponential equations

Answer C

QUESTION 17

The equation $2 \cdot 5^{2x} - 7 \cdot 5^x + 3 = 0$ has two real solutions. What is the sum of these two solutions?

A $\log_5 3$

B $\log_5(1/2)$

C $\log_5(3/2)$

D $\log_5 6$

E $\log_5(2/3)$

WORKED SOLUTION

Let $u = 5^x$. Then $2u^2 - 7u + 3 = 0$, which factors as $(2u-1)(u-3)=0$.

So $u = 1/2$ or $u = 3$.

Hence $x = \log_5(1/2)$ or $x = \log_5 3$.

The sum is $\log_5(1/2) + \log_5 3 = \log_5(3/2)$.

Therefore the answer is **$\log_5(3/2)$** .

M2-18

Coordinate geometry

Answer D

QUESTION 18

The line L with equation $y = mx + c$, where $m > 0$ and $c \geq 0$, passes through the point $(2,4)$. A line is drawn through $(2,4)$ perpendicular to L . The triangle enclosed between the two lines and the y -axis has area 5 square units. What is the larger possible value of m ?

A $\frac{1}{2}$

B 1

C $\frac{3}{2}$

D 2

E $\frac{5}{2}$

WORKED SOLUTION

Since L passes through $(2,4)$, its equation is $y = mx + (4 - 2m)$. So its y -intercept is $4 - 2m$.

The perpendicular line through $(2,4)$ has slope $-1/m$, so its equation is $y - 4 = -(1/m)(x - 2)$.

Its y -intercept is $4 + 2/m$.

The base of the triangle on the y -axis has length $(4 + 2/m) - (4 - 2m) = 2m + 2/m$.

The horizontal distance from $(2,4)$ to the y -axis is 2, so

Area = $\frac{1}{2} \times (2m + 2/m) \times 2 = 2(m + 1/m)$.

Set this equal to 5:

$$2(m + 1/m) = 5 \Rightarrow 2m^2 - 5m + 2 = 0 \Rightarrow (2m-1)(m-2)=0.$$

Thus $m = 1/2$ or 2, and the larger value is **2**.

M2-19

Algebraic structure

Answer **D**

QUESTION 19

How many distinct real solutions does the equation $(x^2 - 2x - 3)^2 = x^2 - 2x - 3$ have?

A 1

B 2

C 3

D 4

E 5

WORKED SOLUTION

Let $u = x^2 - 2x - 3$. Then the equation becomes $u^2 = u$, so $u(u-1)=0$.

Hence $u = 0$ or $u = 1$.

If $u = 0$, then $x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1)=0 \Rightarrow x = 3$ or -1 .

If $u = 1$, then $x^2 - 2x - 4 = 0 \Rightarrow x = 1 \pm \sqrt{5}$.

These are four distinct real solutions, so the answer is **4**.

M2-20

Quadratics

Answer C

QUESTION 20

The curve $y = x^2 - x - 6$ intersects the x -axis at the points A and B, and has a minimum at the point C. A rectangle ABDE has two of its vertices at A and B, and C lies on the edge DE, between D and E. What is the area of the rectangle?

A 25

B $125/8$

C $125/4$

D 35

E 45

WORKED SOLUTION

The roots of $x^2 - x - 6 = 0$ are $x = 3$ and $x = -2$, so $AB = 5$.

The x-coordinate of the minimum is $x = -b/(2a) = 1/2$.

Then y at C is $(1/2)^2 - 1/2 - 6 = 1/4 - 1/2 - 6 = -25/4$.

Since C lies on the top edge DE of the rectangle while A and B lie on the x -axis, the height of the rectangle is $25/4$.

Area = base $\times$ height = $5 \times 25/4 = 125/4$.

M2-21

Trigonometric equations

Answer C

QUESTION 21

It is given that $9\cos x + \tan x \cdot \sin x = 6$, where $0^\circ < x < 90^\circ$. What are the possible values of $\tan x$?

A $\frac{1}{2}$ or $\frac{1}{4}$

B $\frac{1}{\sqrt{3}}$ or $\frac{1}{\sqrt{15}}$

C $\sqrt{3}$ or $\sqrt{15}$

D 2 or 4

E $\sqrt{3}$ or $2\sqrt{2}$

WORKED SOLUTION

Let $c = \cos x$. Then $\tan x \cdot \sin x = \sin^2 x / \cos x = (1 - c^2)/c$.

So the equation becomes $9c + (1 - c^2)/c = 6$.

Multiply by c : $9c^2 + 1 - c^2 = 6c$, so $8c^2 - 6c + 1 = 0$.

Factor: $(4c - 1)(2c - 1) = 0$, giving $c = 1/4$ or $c = 1/2$.

If $\cos x = 1/2$, then $\tan x = \sqrt{3}$.

If $\cos x = 1/4$, then $\sin x = \sqrt{1 - 1/16} = \sqrt{15}/4$, so $\tan x = (\sqrt{15}/4)/(1/4) = \sqrt{15}$.

Therefore the possible values are $\sqrt{3}$ or $\sqrt{15}$.

M2-22

Distance from point to line

Answer **B****QUESTION 22**

A straight line passes through the points $(0, 3a)$ and $(2a, 0)$, where a is a positive constant. What is the perpendicular distance of the point $P(a, 4a)$ from this line?

A $5a/\sqrt{13}$

B $5\sqrt{13} a/13$

C $2\sqrt{13} a/13$

D $4\sqrt{13} a/13$

E $3\sqrt{13} a/13$

WORKED SOLUTION

The line through $(0, 3a)$ and $(2a, 0)$ has intercept form $x/(2a) + y/(3a) = 1$, so $3x + 2y - 6a = 0$.

The perpendicular distance from $(a, 4a)$ to this line is

$$|3a + 2(4a) - 6a| / \sqrt{(3^2 + 2^2)} = |5a|/\sqrt{13} = 5a/\sqrt{13} = 5\sqrt{13} a/13.$$

M2-23

Intersections of curves

Answer **C****QUESTION 23**

At how many distinct points do the curves $y = (x - 3)(x^2 - 4x - 5)$ and $y = -x^2 + 6x - 9$ meet?

A 1

B 2

C 3

D 4

E 5

WORKED SOLUTION

Set the equations equal:

$$(x - 3)(x^2 - 4x - 5) = -x^2 + 6x - 9.$$

Bring everything to one side:

$$(x - 3)(x^2 - 4x - 5) + x^2 - 6x + 9 = 0.$$

Since $x^2 - 6x + 9 = (x-3)^2$, factor out $(x-3)$:

$$(x-3)[x^2 - 4x - 5 + (x-3)] = (x-3)(x^2 - 3x - 8) = 0.$$

So $x = 3$, or $x = (3 \pm \sqrt{41})/2$.

These are three distinct x -values, so the curves meet at **3** distinct points.

M2-24

Geometric progressions

Answer **C**

QUESTION 24

A geometric progression has first term $u_1 = a$ and common ratio r . The sum to infinity is $\frac{8}{5}$, and the sum to infinity of the even-numbered terms ($u_2 + u_4 + u_6 + \dots$) is $\frac{3}{5}$. What is the value of $a + r$?

A $\frac{21}{25}$

B $\frac{6}{5}$

C $\frac{31}{25}$

D $\frac{8}{5}$

E $\frac{41}{25}$

WORKED SOLUTION

For the whole GP, $a/(1-r) = 8/5$.

The even-numbered terms form a GP with first term ar and ratio r^2 , so $ar/(1-r^2) = 3/5$.

But $a = (8/5)(1-r)$. Substitute into the second equation:

$$[(8/5)(1-r)r] / [(1-r)(1+r)] = 3/5.$$

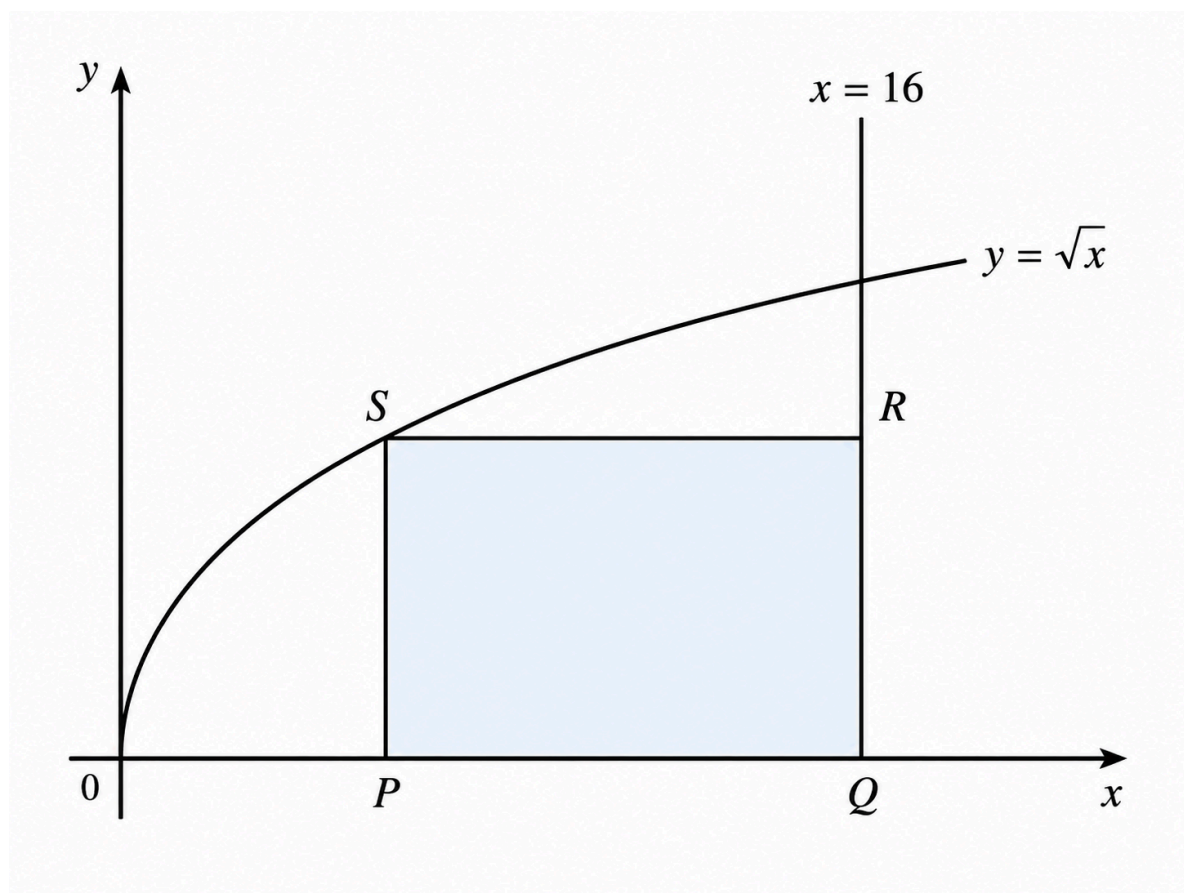
This simplifies to $(8r/5)/(1+r) = 3/5$.

So $8r = 3 + 3r$, hence $5r = 3$ and $r = 3/5$.

Then $a = (8/5)(1-3/5) = (8/5)(2/5) = 16/25$.

Therefore $a + r = 16/25 + 15/25 = \mathbf{31/25}$.

QUESTION 25



PQRS is a rectangle. P and Q lie on the x-axis, Q and R lie on the line $x = 16$, and S lies on the curve $y = \sqrt{x}$. What is the maximum possible area of the rectangle?

A $64\sqrt{3/9}$

B $128\sqrt{3/9}$

C $64\sqrt{3/3}$

D $128\sqrt{3/3}$

E $32\sqrt{3}$

WORKED SOLUTION

Let $S = (x, \sqrt{x})$. Then $P = (x, 0)$, $Q = (16, 0)$, and $R = (16, \sqrt{x})$.

The width of the rectangle is $16 - x$ and its height is $\sqrt{x}$, so

$$A = (16 - x)\sqrt{x}.$$

Let $t = \sqrt{x}$, so $x = t^2$. Then $A = (16 - t^2)t = 16t - t^3$.

Differentiate: $dA/dt = 16 - 3t^2$.

Set to zero: $16 - 3t^2 = 0 \Rightarrow t = 4/\sqrt{3}$.

Then $A_{\max} = 16 \cdot (4/\sqrt{3}) - (4/\sqrt{3})^3 = 64/\sqrt{3} - 64/(3\sqrt{3}) = 128/(3\sqrt{3}) = 128\sqrt{3}/9$.

M2-26

Maximum gradient

Answer E

QUESTION 26

Find the maximum value of the gradient of the curve $y = 1 - 2x + 4x^{3/2} - x^2$, where $x > 0$.

A $\frac{1}{2}$ **B** 1**C** $\frac{3}{2}$ **D** 2**E** $\frac{5}{2}$ **WORKED SOLUTION**

The gradient is $dy/dx = -2 + 6\sqrt{x} - 2x$.

Let $t = \sqrt{x}$, where $t > 0$. Then the gradient becomes

$$g(t) = -2 + 6t - 2t^2 = -2(t^2 - 3t + 1).$$

This is a downward-opening quadratic, so its maximum occurs at $t = -b/(2a) =$

$3/2$.

Then $g_{\max} = -2 + 6 \cdot (3/2) - 2 \cdot (9/4) = -2 + 9 - 9/2 = 5/2$.

M2-27

Trigonometry in triangles

Answer D

QUESTION 27

The internal angles of a triangle are α , β and θ , and

$$3 \tan \alpha - 2 \sin \beta = 2,$$

$$5 \tan \alpha + 6 \sin \beta = 8.$$

What is the value of θ in degrees?

A 15

B 45

C 75

D 105

E 135

WORKED SOLUTION

Let $u = \tan \alpha$ and $v = \sin \beta$. Then we have

$$3u - 2v = 2 \text{ and } 5u + 6v = 8.$$

Multiply the first by 3: $9u - 6v = 6$.

Add to the second: $14u = 14$, so $u = 1$.

Then $3(1) - 2v = 2$, so $v = 1/2$.

Hence $\tan \alpha = 1$. Since α is an internal angle of a triangle, $\alpha = 45^\circ$.

Also $\sin \beta = 1/2$, so an internal angle β could initially be 30° or 150° . But $\beta = 150^\circ$

is impossible because $\alpha + \beta$ would then be 195° , greater than 180° . Thus $\beta = 30^\circ$.
Therefore $\theta = 180^\circ - 45^\circ - 30^\circ = \mathbf{105^\circ}$.

Chemistry

27 QUESTIONS • COMPLETE SOLUTIONS

Preserved Chemistry Harder than ESAT source set. Question wording, option order, answer and solution method are unchanged.

C-01

Atomic structure and isotopes

Answer A

QUESTION 1

An ion X^- contains 18 electrons and 18 neutrons.

A sample of element X contains only two isotopes: the isotope described above and another isotope whose mass number is 2 greater.

The relative atomic mass of X in this sample is 35.6.

What percentage of naturally occurring X atoms are the heavier isotope?

A 30%**B** 20%**C** 25%**D** 35%**E** 40%**F** 60%**WORKED SOLUTION**

Because X^- has 18 electrons, neutral X has 17 electrons and therefore 17 protons.

The stated isotope therefore has mass number

$$17 + 18 = 35.$$

The two isotopes in the sample are 35 and 37. Let the fraction of the heavier isotope be p . Then

$$35(1 - p) + 37p = 35.6.$$

So

$$35 + 2p = 35.6 \Rightarrow p = 0.30.$$

The heavier isotope has abundance 30%, so the answer is **A**.

C-02

Precipitation and dissolved ions

Answer D

QUESTION 2

25.0 cm³ of 0.120 mol dm⁻³ aluminium sulfate solution is mixed with 10.0 cm³ of 0.300 mol dm⁻³ barium chloride solution.

Barium sulfate is insoluble. Assume all other substances remain completely dissociated in solution.

What is the **total amount, in moles, of dissolved ions remaining after precipitation is complete?**

A 0.006

B 0.009

C 0.012

D 0.018

E 0.015

F 0.021

WORKED SOLUTION

Moles of $\text{Al}_2(\text{SO}_4)_3$:

$$0.120 \times 0.0250 = 0.00300.$$

This gives

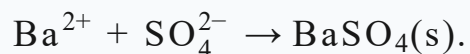
$$0.00600 \text{ mol Al}^{3+}, \quad 0.00900 \text{ mol SO}_4^{2-}.$$

Moles of BaCl_2 :

$$0.300 \times 0.0100 = 0.00300.$$

This gives $0.00300 \text{ mol Ba}^{2+}$ and 0.00600 mol Cl^- .

The precipitation is



All Ba^{2+} is removed, leaving

$$0.00900 - 0.00300 = 0.00600 \text{ mol SO}_4^{2-}.$$

Total dissolved ions remaining:

$$0.00600 + 0.00600 + 0.00600 = \boxed{0.0180 \text{ mol}}.$$

The answer is **D**.

C-03

Structure and bonding

Answer **F**

QUESTION 3

The properties of five substances are shown.

substance	melting point	conducts when solid	conducts when molten	soluble in water
P	high	no	yes	yes
Q	very high	yes	yes	no
R	very high	no	no	no
S	moderate	yes	yes	no
T	very low	no	no	no

Which row gives the most likely structural types of **P**, **Q** and **R**, respectively?

- A** ionic, metallic, simple molecular
- B** giant covalent, metallic, ionic
- C** ionic, ionic, giant covalent
- D** metallic, graphite-like giant covalent, ionic
- E** simple molecular, metallic, giant covalent
- F** ionic, metallic, giant covalent

WORKED SOLUTION

P has the characteristic pattern of an ionic substance: high melting point, no conduction as a solid, conduction when molten and possible water solubility.

Q has a very high melting point, conducts in both the solid and molten states and is insoluble in water, consistent with a metallic structure.

R has a very high melting point and no mobile charge carriers, consistent with a giant covalent structure such as silicon dioxide.

Therefore the answer is **F**.

QUESTION 4

The following pairs of aqueous substances are mixed separately.

1. silver nitrate and potassium bromide
2. chlorine and potassium iodide
3. hydrochloric acid and sodium hydroxide, with no indicator present
4. sodium nitrate and potassium chloride

Which mixtures produce a **visible chemical change**?

A 1 only

B 2 only

C 1 and 3 only

D 2 and 3 only

E 1, 2 and 3 only

F 1, 2 and 4 only

G all four

H 1 and 2 only

WORKED SOLUTION

Mixture 1 forms a precipitate of silver bromide, so there is a visible change.

Mixture 2 undergoes halogen displacement:



giving a colour change.

Mixture 3 undergoes neutralisation, but without an indicator there is no necessary visible change. Mixture 4 leaves all ions in solution.

Therefore **1 and 2 only** are visible, answer **H**.

C-05

Reactivity and extraction

Answer **C**

QUESTION 5

Metal **M** has the following properties.

- It does not react rapidly with cold water.
- It reacts with steam.
- Its oxide can be reduced by carbon.
- It displaces copper from aqueous copper(II) sulfate.

Which could be **M**?

A aluminium

B calcium

C iron

D copper

E magnesium

F potassium

WORKED SOLUTION

The metal must be reactive enough to react with steam and displace copper, but it must lie below carbon because its oxide can be reduced by carbon.

Iron satisfies all of these conditions. Aluminium and magnesium are too reactive for their oxides to be reduced by carbon under the standard extraction model;

calcium and potassium are far more reactive; copper cannot displace itself.

The answer is **C: iron**.

C-06

Oxidising agents

Answer **G**

QUESTION 6

In which reactions is the **first reactant written** acting as an oxidising agent?

1. $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$
2. $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$
3. $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$
4. $2\text{Fe}^{3+} + 2\text{I}^- \rightarrow 2\text{Fe}^{2+} + \text{I}_2$

A 1 only

B 1 and 3 only

C 1 and 4 only

D 2 and 3 only

E 2, 3 and 4 only

F all four

G 1, 3 and 4 only

WORKED SOLUTION

An oxidising agent is itself reduced.

1. Chlorine changes from oxidation state 0 to -1 , so Cl_2 is reduced and is an oxidising agent.

2. Zinc changes from 0 to +2, so it is oxidised and acts as a reducing agent.
3. Manganese changes from +4 in MnO_2 to +2 in MnCl_2 , so MnO_2 is reduced.
4. $\text{Fe}^{3+} \rightarrow \text{Fe}^{2+}$ is reduction.

Thus the first reactant is an oxidising agent in **1, 3 and 4**, answer **G**.

C-07

Balancing an unfamiliar redox reaction

Answer **B**

QUESTION 7

A divalent metal M reacts with nitric acid to form:

- $\text{M}(\text{NO}_3)_2$,
- water,
- a single oxide of nitrogen X .

Exactly 0.075 mol of M reacts with 0.300 mol of nitric acid. No other products are formed.

What is the empirical formula of X ?

A N_2O

B NO_2

C NO

D N_2O_3

E N_2O_5

WORKED SOLUTION

0.075 mol of M forms 0.075 mol of $\text{M}(\text{NO}_3)_2$, using

$$2(0.075) = 0.150 \text{ mol N atoms}$$

in nitrate.

Total nitrogen supplied is 0.300 mol, so nitrogen in X is

$$0.300 - 0.150 = 0.150 \text{ mol N atoms.}$$

The 0.300 mol of nitric acid supplies 0.300 mol of H atoms, so 0.150 mol of water is formed.

Total oxygen supplied is

$$3(0.300) = 0.900 \text{ mol O atoms.}$$

Oxygen used in nitrate is $6(0.075) = 0.450$ mol, and oxygen used in water is 0.150 mol. Therefore oxygen in X is

$$0.900 - 0.450 - 0.150 = 0.300.$$

Thus

$$\text{N} : \text{O} = 0.150 : 0.300 = 1 : 2,$$

so $X = \boxed{\text{NO}_2}$. The answer is **B**.

C-08

Acid-base mixture and pH

Answer **G**

QUESTION 8

50.0 cm³ of 0.100 mol dm⁻³ hydrochloric acid is mixed with 150 cm³ of 0.0125 mol dm⁻³ calcium hydroxide solution.

Which range contains the pH of the final solution?

A $0 < \text{pH} < 1$

B $1 < \text{pH} < 2$

C $3 < \text{pH} < 7$

D $\text{pH} = 7$

E $7 < \text{pH} < 12$

F $12 < \text{pH} < 14$

G $2 < \text{pH} < 3$

WORKED SOLUTION

Moles of H^+ :

$$0.100(0.0500) = 0.00500.$$

Moles of $\text{Ca}(\text{OH})_2$:

$$0.0125(0.150) = 0.001875.$$

This supplies

$$2(0.001875) = 0.00375 \text{ mol OH}^-.$$

Hydrogen ions are in excess:

$$0.00500 - 0.00375 = 0.00125 \text{ mol H}^+.$$

Total volume is 0.200 dm^3 , so

$$[\text{H}^+] = \frac{0.00125}{0.200} = 0.00625 \text{ mol dm}^{-3}.$$

This concentration lies between 10^{-3} and $10^{-2} \text{ mol dm}^{-3}$. Therefore the pH lies between 2 and 3:

$$\boxed{2 < \text{pH} < 3}.$$

The answer is **G**.

QUESTION 9

An acid solution has

$$\text{concentration} = 0.300 \text{ mol dm}^{-3}$$

and

$$\text{mass concentration} = 29.4 \text{ g dm}^{-3}.$$

A 25.0 cm^3 sample is completely neutralised by 45.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ sodium hydroxide. The acid also reacts with magnesium to produce hydrogen.

Which statements can be deduced?

1. The relative molecular mass of the acid is 98.
2. One mole of the acid can supply three moles of H^+ in complete neutralisation.
3. The acid is a strong acid.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The molar mass is

$$M_r = \frac{29.4}{0.300} = 98,$$

so statement 1 is true.

Moles of acid in 25.0 cm^3 :

$$0.300(0.0250) = 0.00750.$$

Moles of NaOH:

$$0.500(0.0450) = 0.0225.$$

The ratio is

$$\frac{0.0225}{0.00750} = 3,$$

so statement 2 is true.

Reaction with magnesium does not show whether the acid is fully ionised in water, so statement 3 cannot be deduced.

The answer is **E: 1 and 2 only**.

C-10

Back titration and purity

Answer **F**

QUESTION 10

A 2.10 g impure sample of magnesium carbonate contains impurities that do not react with acids.

The sample is reacted with 100 cm^3 of $0.500 \text{ mol dm}^{-3}$ hydrochloric acid.

The excess hydrochloric acid then requires 20.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ sodium hydroxide for complete neutralisation.



$$M_r(\text{MgCO}_3) = 84$$

What is the percentage by mass of magnesium carbonate in the original sample?

A 40%

B 50%

C 60%

D 70%

E 90%

F 80%

WORKED SOLUTION

Initial HCl:

$$0.500(0.100) = 0.0500 \text{ mol.}$$

HCl remaining is equal to the amount of NaOH used:

$$0.500(0.0200) = 0.0100 \text{ mol.}$$

Therefore HCl consumed by MgCO_3 is

$$0.0500 - 0.0100 = 0.0400 \text{ mol.}$$

From the 1:2 ratio,

$$n(\text{MgCO}_3) = 0.0200 \text{ mol.}$$

Mass of pure MgCO_3 :

$$0.0200(84) = 1.68 \text{ g.}$$

Hence

$$\% \text{ purity} = \frac{1.68}{2.10} \times 100 = \boxed{80\%}$$

The answer is **F**.

C-11

Molecular formula from combustion

Answer **C**

QUESTION 11

A compound contains carbon, hydrogen and oxygen only.

Complete combustion of 0.880 g of the compound produces 1.760 g of CO_2 and 0.720 g of H_2O .

The relative molecular mass of the compound is 88.

$$A_r : \quad \text{C} = 12, \text{H} = 1, \text{O} = 16$$

What is the molecular formula?

A CH_2O

B $\text{C}_2\text{H}_4\text{O}$

C $\text{C}_4\text{H}_8\text{O}_2$

D $\text{C}_2\text{H}_4\text{O}_2$

E $\text{C}_3\text{H}_6\text{O}_2$

F $\text{C}_4\text{H}_8\text{O}$

WORKED SOLUTION

Carbon:

$$n(\text{CO}_2) = \frac{1.760}{44} = 0.0400,$$

so there are 0.0400 mol C atoms.

Hydrogen:

$$n(\text{H}_2\text{O}) = \frac{0.720}{18} = 0.0400,$$

so there are 0.0800 mol H atoms.

Mass of carbon is $0.0400(12) = 0.480$ g; mass of hydrogen is 0.0800 g.

Therefore oxygen mass is

$$0.880 - 0.480 - 0.080 = 0.320 \text{ g},$$

giving

$$n(\text{O}) = \frac{0.320}{16} = 0.0200.$$

The ratio is

$$\text{C} : \text{H} : \text{O} = 0.040 : 0.080 : 0.020 = 2 : 4 : 1.$$

Empirical formula: $\text{C}_2\text{H}_4\text{O}$, mass 44. Since $88 = 2(44)$, the molecular formula is



The answer is **C**.

C-12

Composition of air

Answer **B**

QUESTION 12

A sample of dry air contains exactly 30.0 mol of gas.

Gas X contributes 3.78×10^{24} molecules to the sample.

Gas Y, if isolated at room temperature and pressure, would occupy 561.6 dm^3 .

Take $N_A = 6.00 \times 10^{23}$ and molar gas volume at rtp as $24.0 \text{ dm}^3 \text{ mol}^{-1}$.

Which row is consistent with the approximate composition of dry air?

option	X	Y	remaining gases / mol
A	N ₂	O ₂	0.30
B	O ₂	N ₂	0.30
C	Ar	N ₂	6.30
D	O ₂	Ar	23.4
E	N ₂	Ar	6.30

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

For X:

$$n = \frac{3.78 \times 10^{24}}{6.00 \times 10^{23}} = 6.30 \text{ mol.}$$

That is

$$\frac{6.30}{30.0} = 0.21 = 21\%,$$

so X is oxygen.

For Y:

$$n = \frac{561.6}{24.0} = 23.4 \text{ mol,}$$

which is

$$\frac{23.4}{30.0} = 0.78 = 78\%,$$

so Y is nitrogen.

Remaining gases:

$$30.0 - 6.30 - 23.4 = 0.30 \text{ mol.}$$

Therefore the correct row is **B**.

C-13

Two-dimensional chromatography

Answer **E**

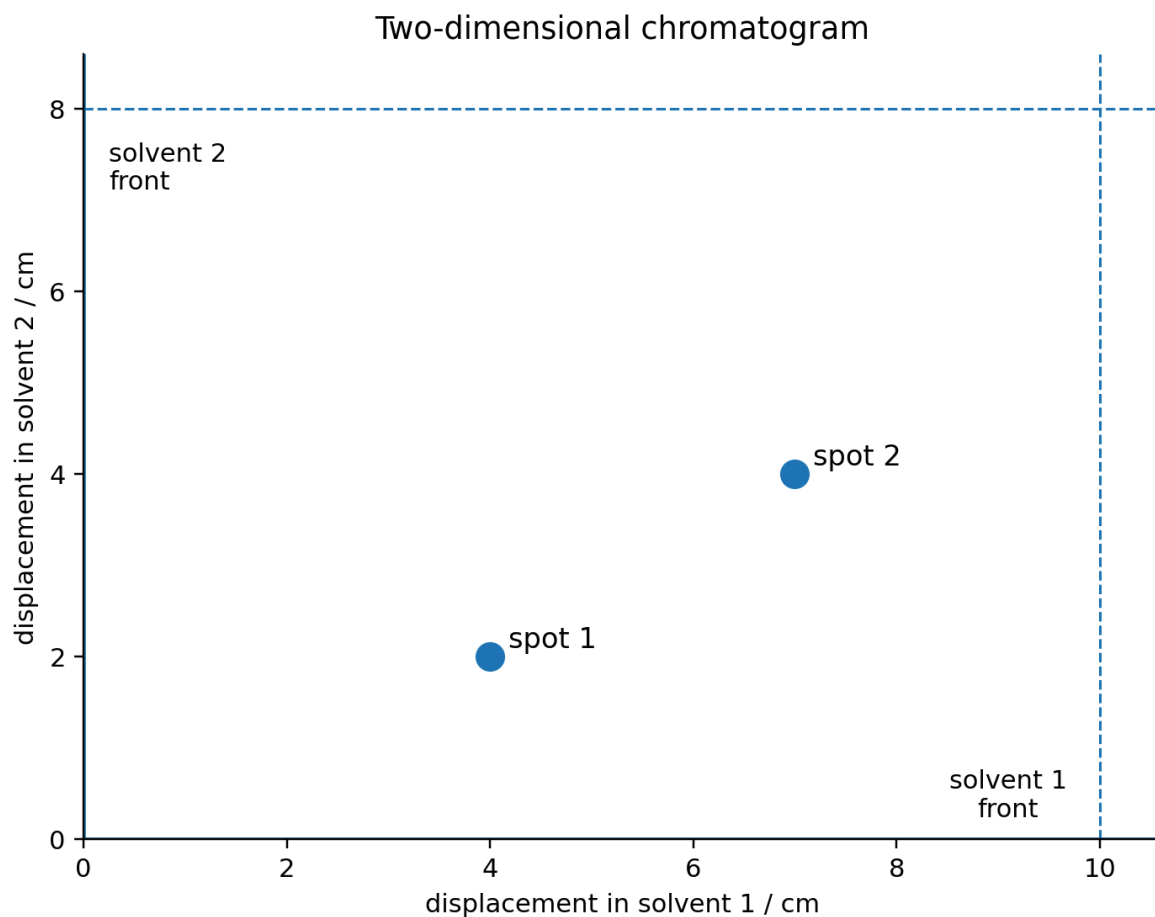
QUESTION 13

Four compounds have the following R_f values in two solvents.

compound	R_f in solvent 1	R_f in solvent 2
A	0.20	0.75
B	0.40	0.25
C	0.40	0.75
D	0.70	0.50

A mixture is analysed by two-dimensional chromatography. In the first direction, solvent 1 travels 10.0 cm. The paper is dried, rotated through 90° , and solvent 2 travels 8.0 cm.

The final chromatogram is shown.



Which compounds were present?

A A and B

B A and C

C B and C

D C and D

E B and D

WORKED SOLUTION

For spot 1:

$$R_{f,1} = \frac{4.0}{10.0} = 0.40, \quad R_{f,2} = \frac{2.0}{8.0} = 0.25.$$

This identifies compound B.

For spot 2:

$$R_{f,1} = \frac{7.0}{10.0} = 0.70, \quad R_{f,2} = \frac{4.0}{8.0} = 0.50.$$

This identifies compound D.

Therefore the mixture contained **B and D**, answer **E**.

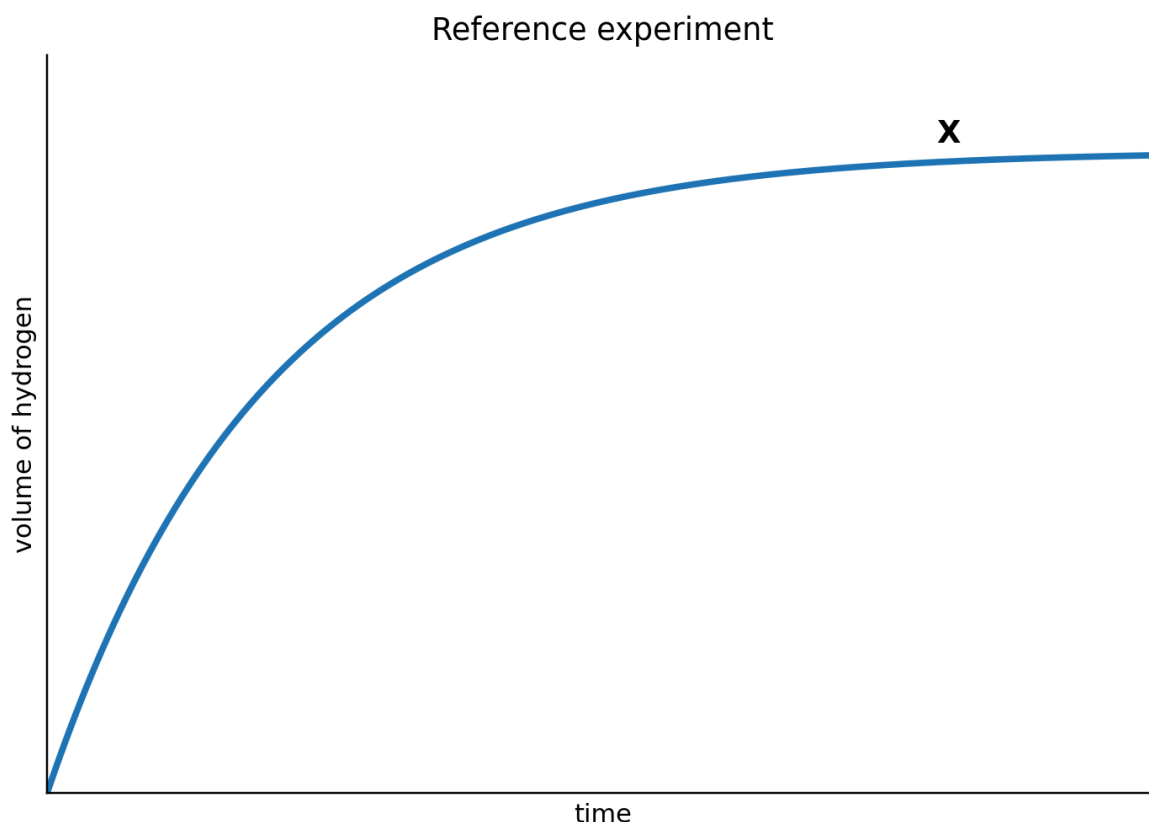
C-14

Rate and limiting reagent

Answer A

QUESTION 14

Curve X is obtained when 0.240 g of magnesium granules react with 100 cm³ of 0.100 mol dm⁻³ hydrochloric acid.



The experiment is repeated using:

- the same mass of magnesium, but powdered;
- 100 cm³ of 0.150 mol dm⁻³ hydrochloric acid.

$$A_r(\text{Mg}) = 24$$

Which statement correctly compares the second curve with X?

option	initial gradient	final volume of H_2
A	greater	$1.5 \times X$
B	greater	same as X
C	same	$1.5 \times X$
D	smaller	$1.5 \times X$
E	greater	$2.0 \times X$

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

Moles of magnesium:

$$\frac{0.240}{24} = 0.0100.$$

Complete reaction would need 0.0200 mol HCl.

First experiment:

$$n(\text{HCl}) = 0.100(0.100) = 0.0100 \text{ mol},$$

so acid is limiting.

Second experiment:

$$n(\text{HCl}) = 0.150(0.100) = 0.0150 \text{ mol},$$

which is still limiting. Therefore the final hydrogen amount is larger by

$$\frac{0.0150}{0.0100} = 1.5.$$

The second experiment also has a higher acid concentration and a larger magnesium surface area, so its initial rate is greater.

The correct row is **A**.

C-15

Equilibrium from observations

Answer C

QUESTION 15

Gaseous reactants P and Q establish equilibrium with gaseous products R and S.

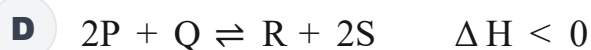
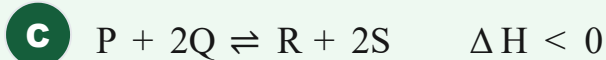
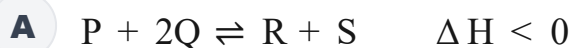
During establishment of equilibrium from the reactants:

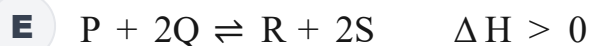
- for every 0.10 mol of P consumed,
- 0.20 mol of Q is consumed,
- 0.10 mol of R is formed.

Increasing pressure changes the rate at which equilibrium is reached but does **not** change the equilibrium yield.

Increasing temperature reduces the equilibrium yield of products.

Which equation is consistent with all the observations?



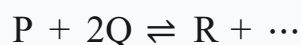


WORKED SOLUTION

The observed mole changes give the ratio

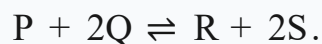
$$P : Q : R = 1 : 2 : 1.$$

So the equation must begin



There are three gaseous reactant particles. Because pressure does not shift the equilibrium yield, the product side must also contain three gaseous particles.

Thus the stoichiometry is



Increasing temperature reduces product yield, so the forward reaction is exothermic: $\Delta H < 0$.

The answer is **C**.

C-16

Bond energies

Answer **D**

QUESTION 16

Ethene reacts with hydrogen:



The enthalpy change is -125 kJ mol^{-1} .

Mean bond energies are:

bond	bond energy / kJ mol^{-1}
H–H	436
C=C	612
C–C	348

What is the mean C–H bond energy implied by these data?

A 356 kJ mol^{-1}

B 388 kJ mol^{-1}

C 436 kJ mol^{-1}

D 413 kJ mol^{-1}

E 475 kJ mol^{-1}

WORKED SOLUTION

Relative to the reactants, hydrogenation breaks one C=C bond and one H–H bond, and forms one C–C bond and two additional C–H bonds.

Let the mean C–H bond energy be x . Then

$$-125 = (612 + 436) - (348 + 2x).$$

So

$$-125 = 700 - 2x \Rightarrow 2x = 825 \Rightarrow x = 412.5.$$

Therefore the closest value is 413 kJ mol^{-1} , answer **D**.

QUESTION 17

40.0 cm³ of 1.50 mol dm⁻³ hydrochloric acid is mixed with 60.0 cm³ of 0.500 mol dm⁻³ sodium hydroxide.

The temperature rises by 3.4°C.

Assume:

- the solution volumes are additive;
- density of the final solution = 1.0 g cm⁻³;
- specific heat capacity = 4.2 J g⁻¹ °C⁻¹;
- only 85% of the energy released by the reaction is retained by the solution.

What is the molar enthalpy change for the neutralisation reaction?

A - 28 kJ mol⁻¹

B - 40 kJ mol⁻¹

C - 56 kJ mol⁻¹

D - 48 kJ mol⁻¹

E - 66 kJ mol⁻¹

WORKED SOLUTION

Moles HCl:

$$1.50(0.0400) = 0.0600.$$

Moles NaOH:

$$0.500(0.0600) = 0.0300.$$

NaOH is limiting, so 0.0300 mol is neutralised.

The solution mass is 100 g. Energy retained:

$$q = mc\Delta T = 100(4.2)(3.4) = 1428 \text{ J.}$$

This is 85% of the energy released, so

$$q_{\text{released}} = \frac{1428}{0.85} = 1680 \text{ J} = 1.680 \text{ kJ.}$$

Per mole:

$$\frac{1.680}{0.0300} = 56.0 \text{ kJ mol}^{-1}.$$

Neutralisation is exothermic, so $\boxed{-56.0 \text{ kJ mol}^{-1}}$, answer **C**.

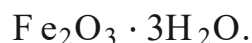
C-18

Metal extraction

Answer **B**

QUESTION 18

An iron ore contains 80.0% by mass of hydrated iron(III) oxide,



The formula $\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$ means that each formula unit contains one Fe_2O_3 unit together with three H_2O units.

Assume all the iron in the hydrated oxide can be extracted.

$$A_r : \quad \text{Fe} = 56, \text{O} = 16, \text{H} = 1$$

What is the minimum mass of ore required to produce 56.0 tonnes of iron?

A 100 tonnes

B 134 tonnes

C 112 tonnes

D 125 tonnes

E 160 tonnes

F 214 tonnes

WORKED SOLUTION

The relative formula mass of $\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$ is

$$2(56) + 3(16) + 3(18) = 214.$$

Each 214 mass units contain 112 mass units of iron, so the iron fraction in pure hydrated oxide is $112/214$.

The ore is only 80.0% hydrated oxide, so the iron fraction of the ore is

$$0.80 \left(\frac{112}{214} \right).$$

If m tonnes of ore are needed,

$$m \cdot 0.80 \left(\frac{112}{214} \right) = 56.0.$$

Thus

$$m = 133.75 \text{ tonnes} \approx \boxed{134 \text{ tonnes}}.$$

The answer is **B**.

C-19

Electrolysis and concentration

Answer **E**

QUESTION 19

250 cm^3 of 1.00 mol dm^{-3} copper(II) sulfate solution is electrolysed using inert electrodes.

After some time, 6.40 g of copper has been deposited at the cathode.

Water is then added so that the solution volume is again exactly 250 cm^3 .

$$A_r(\text{Cu}) = 64$$

What is the final concentration of Cu^{2+} ?

A 0.20 mol dm^{-3}

B 0.40 mol dm^{-3}

C 0.50 mol dm^{-3}

D 0.75 mol dm^{-3}

E 0.60 mol dm^{-3}

F 0.90 mol dm^{-3}

WORKED SOLUTION

Initial amount of Cu^{2+} :

$$1.00(0.250) = 0.250\text{ mol.}$$

Copper deposited:

$$n(\text{Cu}) = \frac{6.40}{64} = 0.100\text{ mol.}$$

Therefore 0.100 mol of Cu^{2+} has been removed, leaving

$$0.250 - 0.100 = 0.150\text{ mol.}$$

At a final volume of 0.250 dm^3 ,

$$[\text{Cu}^{2+}] = \frac{0.150}{0.250} = \boxed{0.600\text{ mol dm}^{-3}}.$$

The answer is **E**.

C-20

Organic structure from constraints

Answer **B**

QUESTION 20

A hydrocarbon X:

- produces 120 dm^3 of carbon dioxide when one mole is completely burned;
- reacts with exactly one mole of hydrogen per mole of X;
- undergoes addition polymerisation;
- has a terminal C=C bond;
- forms a **branched saturated hydrocarbon** on hydrogenation.

At rtp, one mole of gas occupies 24 dm^3 .

The candidate structures are shown as condensed structural formulae.

structure	condensed structural formula
A	$\text{CH}_2 = \text{CHCH}_2\text{CH}_2\text{CH}_3$
B	$\text{CH}_2 = \text{C}(\text{CH}_3)\text{CH}_2\text{CH}_3$
C	$\text{CH}_3\text{CH} = \text{CHCH}_2\text{CH}_3$
D	$\text{CH}_3\text{C}(\text{CH}_3) = \text{CHCH}_3$
E	cyclopentane, $(\text{CH}_2)_5$ ring

Which could be X?

A structure A

B structure B

C structure C

D structure D

E structure E

WORKED SOLUTION

The carbon dioxide volume corresponds to

$$\frac{120}{24} = 5 \text{ mol CO}_2,$$

so X contains five carbon atoms.

Reaction with exactly one mole of hydrogen indicates one C=C bond. Addition polymerisation also requires an alkene.

The double bond must be terminal, leaving A or B. Hydrogenating A gives unbranched pentane, whereas hydrogenating B gives branched 2-methylbutane.

Therefore X is **structure B**.

C-21

Multiple double bonds

Answer A

QUESTION 21

A hydrocarbon contains two carbon-carbon double bonds.

A 0.100 cm^3 sample has density 0.820 g cm^{-3} and relative molecular mass 82.

What is the minimum volume of $0.250 \text{ mol dm}^{-3}$ bromine solution required to react completely with the hydrocarbon?

A 8.0 cm^3

B 2.0 cm^3

C 4.0 cm^3

D 6.0 cm^3

E 16.0 cm³

WORKED SOLUTION

Mass of hydrocarbon:

$$0.100(0.820) = 0.0820 \text{ g.}$$

Moles:

$$\frac{0.0820}{82} = 0.00100 \text{ mol.}$$

There are two C=C bonds, so each mole consumes two moles of bromine. Thus

$$n(\text{Br}_2) = 0.00200 \text{ mol.}$$

Required volume:

$$V = \frac{0.00200}{0.250} = 0.00800 \text{ dm}^3 = \boxed{8.00 \text{ cm}^3}.$$

The answer is **A**.

C-22

Carboxylic acid and esterification

Answer **D**

QUESTION 22

A saturated monocarboxylic acid X reacts with propan-1-ol to form an ester Y.

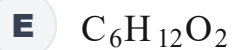
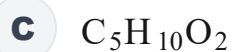
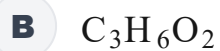
$$M_r(\text{Y}) = 130$$

and

$$M_r(\text{propan-1-ol}) = 60.$$

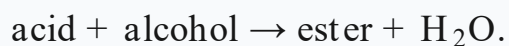
What is the molecular formula of X?

A C₂H₄O₂



WORKED SOLUTION

Esterification gives



Therefore

$$M_r(\text{X}) + 60 = 130 + 18,$$

so

$$M_r(\text{X}) = 88.$$

A saturated monocarboxylic acid has formula $\text{C}_n\text{H}_{2n}\text{O}_2$, so

$$12n + 2n + 32 = 88.$$

Thus

$$14n = 56 \Rightarrow n = 4.$$

The formula is $\boxed{\text{C}_4\text{H}_8\text{O}_2}$, answer **D**.

C-23

Combustion and cracking

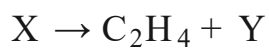
Answer **F**

QUESTION 23

20 cm^3 of a gaseous alkane **X** requires 190 cm^3 of oxygen for complete combustion.

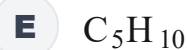
Both gas volumes are measured at the same temperature and pressure.

X can crack according to



with one molecule of Y formed for each molecule of X.

What is Y?



WORKED SOLUTION

For an alkane C_nH_{2n+2} , complete combustion requires

$$n + \frac{2n + 2}{4} = \frac{3n + 1}{2}$$

moles of oxygen per mole of alkane.

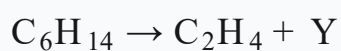
The gas-volume ratio is

$$\frac{190}{20} = 9.5.$$

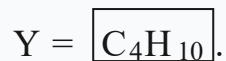
Hence

$$\frac{3n + 1}{2} = 9.5 \Rightarrow 3n + 1 = 19 \Rightarrow n = 6.$$

So $X = C_6H_{14}$. Balancing



gives



The answer is **F**.

C-24

Water treatment and gas quantities

Answer **D**

QUESTION 24

A water-treatment plant processes $480\,000\text{ dm}^3$ of water each day.

Chlorine is added at a rate of 0.355 mg dm^{-3} .

Assume all the chlorine is supplied as $\text{Cl}_2(\text{g})$.

$$A_r(\text{Cl}) = 35.5$$

and one mole of gas occupies 24.0 dm^3 at rtp.

What volume of chlorine gas is used each day?

A 24.0 dm^3

B 36.0 dm^3

C 48.0 dm^3

D 57.6 dm^3

E 71.0 dm^3

F 96.0 dm^3

WORKED SOLUTION

Mass of chlorine used:

$$480000(0.355) = 170400 \text{ mg} = 170.4 \text{ g.}$$

Since $M_r(\text{Cl}_2) = 71$,

$$n(\text{Cl}_2) = \frac{170.4}{71} = 2.40 \text{ mol.}$$

At rtp,

$$V = 2.40(24.0) = \boxed{57.6 \text{ dm}^3}.$$

The answer is **D**.

C-25

Qualitative analysis

Answer **F**

QUESTION 25

An aqueous salt **X** gives:

- a **green precipitate** on addition of aqueous sodium hydroxide;
- a **white precipitate** on addition of acidified aqueous barium chloride.

Which is the most likely identity of **X**?

A FeCl_2

B $\text{Fe}_2(\text{SO}_4)_3$

C CuSO_4

D MgSO_4

E Na_2SO_4

F FeSO_4

WORKED SOLUTION

A green hydroxide precipitate identifies Fe^{2+} .

A white precipitate with acidified barium chloride identifies sulfate ions, SO_4^{2-} .

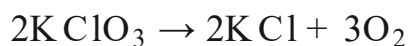
The salt must therefore be FeSO_4 , answer **F**.

C-26

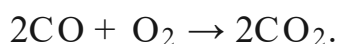
Multi-stage gas stoichiometry

Answer **E****QUESTION 26**

Potassium chlorate decomposes completely:



All the oxygen produced is then used to oxidise excess carbon monoxide:



A 24.5 g sample of pure potassium chlorate is used.

$$M_r(\text{KClO}_3) = 122.5$$

and one mole of gas occupies 24.0 dm^3 at rtp.

What volume of carbon dioxide is ultimately formed?

A 4.8 dm^3

B 7.2 dm^3

C 9.6 dm^3

D 12.0 dm^3

E 14.4 dm^3

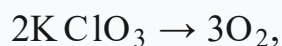
F 18.0 dm³

WORKED SOLUTION

Moles of potassium chlorate:

$$\frac{24.5}{122.5} = 0.200 \text{ mol.}$$

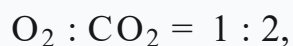
From



the oxygen amount is

$$0.200 \left(\frac{3}{2} \right) = 0.300 \text{ mol.}$$

Then



so

$$n(\text{CO}_2) = 0.600 \text{ mol.}$$

Volume:

$$0.600(24.0) = \boxed{14.4 \text{ dm}^3}.$$

The answer is **E**.

C-27

Mixture analysis and back titration

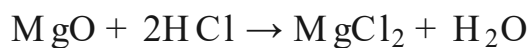
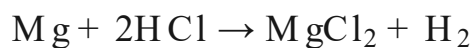
Answer **A**

QUESTION 27

A 3.00 g mixture contains only magnesium and magnesium oxide.

It is reacted with 100 cm³ of 2.00 mol dm⁻³ hydrochloric acid.

The excess hydrochloric acid requires 20.0 cm³ of 1.00 mol dm⁻³ sodium hydroxide for complete neutralisation.



$$A_r(\text{Mg}) = 24, \quad M_r(\text{MgO}) = 40.$$

What volume of hydrogen gas is produced at rtp?

$$1 \text{ mol gas} = 24.0 \text{ dm}^3.$$

A 0.90 dm^3

B 0.45 dm^3

C 0.60 dm^3

D 0.75 dm^3

E 1.20 dm^3

F 1.80 dm^3

WORKED SOLUTION

Initial HCl:

$$0.100(2.00) = 0.200 \text{ mol.}$$

Excess HCl:

$$0.0200(1.00) = 0.0200 \text{ mol.}$$

So HCl consumed by the mixture is

$$0.200 - 0.0200 = 0.180 \text{ mol.}$$

Both Mg and MgO consume two moles HCl per mole, so

$$n(\text{Mg}) + n(\text{MgO}) = \frac{0.180}{2} = 0.0900.$$

Let $n(\text{Mg}) = x$ and $n(\text{MgO}) = y$. Then

$$x + y = 0.0900$$

and

$$24x + 40y = 3.00.$$

Using $x = 0.0900 - y$:

$$24(0.0900 - y) + 40y = 3.00$$

$$2.16 + 16y = 3.00 \Rightarrow y = 0.0525.$$

Hence

$$x = 0.0900 - 0.0525 = 0.0375.$$

Only magnesium metal produces hydrogen, in a 1:1 ratio, so

$$n(\text{H}_2) = 0.0375.$$

Therefore

$$V(\text{H}_2) = 0.0375(24.0) = \boxed{0.900 \text{ dm}^3}.$$

The answer is **A**.

Compilation record: Level 2 uses Engineering Mock Test 5 for Mathematics/Physics and the corresponding Harder than ESAT Biology/Chemistry source sets. This solution book is ready to support the later construction of the matching full test.